

kohlhase-omdoc: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00001	10	■ 1.000	\Omega	Ω	tably the Ω MEGA group at Saarland University, the MAYA and ACTIVEMATH
kohlhase-omdoc_F00002	19	■ 0.928	\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{A}_{\mathrm{E}} \mathrm{X}	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{A}_{\mathrm{E}} \mathrm{X}$	distinguish them from presentation markup schemes, and discuss $\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{A}_{\mathrm{E}} \mathrm{X}$
kohlhase-omdoc_F00003	19	■ 1.000	\mathrm{T}_{\mathrm{E}} \mathrm{X}	$\mathrm{T}_{\mathrm{E}} \mathrm{X}$	$\mathrm{T}_{\mathrm{E}} \mathrm{X}$ is a typesetting markup language that uses explicit markup codes
kohlhase-omdoc_F00004	19	—	\sqrt{\sin x}	(not rendered)	—
kohlhase-omdoc_F00005	19	■ 0.878	\sqrt{\sin x}	$\sqrt{\sin x}$	$\sqrt{\sin x}$ stands for the mathematical expression $\sqrt{\sin x}$ in $\mathrm{T}_{\mathrm{E}} \mathrm{X}$. To
kohlhase-omdoc_F00006	21	■ 0.696	\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{a} \mathrm{T}_{\mathrm{E}} \mathrm{X}	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{a} \mathrm{T}_{\mathrm{E}} \mathrm{X}$	to $\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{a} \mathrm{T}_{\mathrm{E}} \mathrm{X}$ in spirit, but tries to give the markup scheme a more declar-
kohlhase-omdoc_F00007	23	■ 0.940	\langle \! \langle nsURI \rangle \! \rangle	$\langle \! \langle nsURI \rangle \! \rangle$	descendants are in the namespace $\langle \! \langle nsURI \rangle \! \rangle$, unless they have a namespace
kohlhase-omdoc_F00008	23	■ 0.198	\langle	$\langle$	by a pseudo-attribute of the form $\mathrm{xm} \mathrm{l} \mathrm{n} \mathrm{s} \! : \! \langle \! \langle ns \rangle \! \rangle = \! \langle \! \langle nsUR \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is
kohlhase-omdoc_F00009	23	■ 0.198	\rangle =	$\rangle =$	by a pseudo-attribute of the form $\mathrm{xm} \mathrm{l} \mathrm{n} \mathrm{s} \! : \! \langle \! \langle ns \rangle \! \rangle = \! \langle \! \langle nsUR \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is
kohlhase-omdoc_F00010	23	■ 0.198	\rangle	$\rangle$	by a pseudo-attribute of the form $\mathrm{xm} \mathrm{l} \mathrm{n} \mathrm{s} \! : \! \langle \! \langle ns \rangle \! \rangle = \! \langle \! \langle nsUR \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is
kohlhase-omdoc_F00011	23	■ 0.198	\langle \! \langle	$\langle \! \langle$	by a pseudo-attribute of the form $\mathrm{xm} \mathrm{l} \mathrm{n} \mathrm{s} \! : \! \langle \! \langle ns \rangle \! \rangle = \! \langle \! \langle nsUR \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is
kohlhase-omdoc_F00012	23	■ 0.198	\rangle \! \rangle	$\rangle \! \rangle$	by a pseudo-attribute of the form $\mathrm{xm} \mathrm{l} \mathrm{n} \mathrm{s} \! : \! \langle \! \langle ns \rangle \! \rangle = \! \langle \! \langle nsUR \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is
kohlhase-omdoc_F00013	23	■ 0.190	n s a \rangle	$nsa \rangle \! \rangle$	a qualified name : a pair $\langle \! \langle ns \rangle \! \rangle \! : \! \langle \! \langle el \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is a namespace abbrevi-
kohlhase-omdoc_F00014	23	■ 0.190	n s a \rangle	$nsa \rangle$	a qualified name : a pair $\langle \! \langle ns \rangle \! \rangle \! : \! \langle \! \langle el \rangle \! \rangle$, where $\langle \! \langle ns \rangle \! \rangle$ is a namespace abbrevi-
kohlhase-omdoc_F00015	23	■ 0.917	\hat{=}	$\hat{=}$	$\langle \! \langle CDATA[a < b < \sup > 3 < / \sup >] \rangle \! \rangle \hat{=} a \< b \< \sup \> 3 \< / \sup \>$
kohlhase-omdoc_F00016	25	■ 0.832	7^{\text{th}}	7^{th}	the 7^{th} child of the 3^{rd} child of the (unique) element that has ID-type attribute
kohlhase-omdoc_F00017	25	■ 0.832	3^{\text{rd}}	3^{rd}	the 7^{th} child of the 3^{rd} child of the (unique) element that has ID-type attribute
kohlhase-omdoc_F00018	25	■ 0.984	D	D	(the shorthand xpointer): If $\langle \! \langle uri \rangle \! \rangle$ is a URI of an XML document D then
kohlhase-omdoc_F00019	25	■ 0.454	\langle \! \langle id \rangle \! \rangle	$\langle \! \langle id \rangle \! \rangle$	ID with value $\langle \! \langle id \rangle \! \rangle$.
kohlhase-omdoc_F00020	27	■ 0.990	\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{A}_{\mathrm{T}} \mathrm{E} \mathrm{X}	$\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{A}_{\mathrm{T}} \mathrm{E} \mathrm{X}$	systems like $\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{L} \mathrm{A}_{\mathrm{T}} \mathrm{E} \mathrm{X}$.
kohlhase-omdoc_F00021	28	■ 0.348	{ }^{+}	$+$	formulae for the Web are MATHML [ABC ⁺ 03] and OPENMATH [BCC ⁺ 04].
kohlhase-omdoc_F00022	28	■ 1.000	{ }^{\text{®}}	®	mathematical software packages like Wolfram Research’s MATHEMATICA [®]
kohlhase-omdoc_F00023	29	■ 1.000	\frac{3}{x+2}	$\frac{3}{x+2}$	the expression $\frac{3}{x+2}$ would be represented as the layout tree in Figure 2.2.

Identifier	Page	Conf.	LaTeX source	Rendered	Image								
kohlhase-omdoc_F00024	29	■ 0.991	\Sigma	Σ	DTD, e.g. ∑ for Σ .								
kohlhase-omdoc_F00025	29	■ 0.840	14{ }^{\text{th}}	14 th	fixed set of mathematical concepts – the K-14 fragment (Kindergarten to 14 th								
kohlhase-omdoc_F00026	30	■ 0.998	\mathrm{T}_{\mathrm{E}}\mathrm{X}^3	$\mathrm{T}_{\mathrm{E}}\mathrm{X}^3$	of $\mathrm{T}_{\mathrm{E}}\mathrm{X}$. However, it does not offer the programming language features of $\mathrm{T}_{\mathrm{E}}\mathrm{X}^3$								
kohlhase-omdoc_F00027	30	■ 1.000	{ }^3 \mathrm{T}_{\mathrm{E}}\mathrm{X}	${}^3 \mathrm{T}_{\mathrm{E}}\mathrm{X}$	${}^3 \mathrm{T}_{\mathrm{E}}\mathrm{X}$ contains a full, Turing-complete – if somewhat awkward – programming lan-								
kohlhase-omdoc_F00028	32	■ 0.992	\forall a, b . a+b=b+a	$\forall a, b. a + b = b + a$	Listing 2.5. OPENMATH representation of $\forall a, b. a + b = b + a$								
kohlhase-omdoc_F00029	35	■ 1.000	S=(G, \circ)	$S = (G, \circ)$	A monoid is a semigroup $S = (G, \circ)$ with an element $e \in G$, such that								
kohlhase-omdoc_F00030	35	■ 1.000	e \in G	$e \in G$	A monoid is a semigroup $S = (G, \circ)$ with an element $e \in G$, such that								
kohlhase-omdoc_F00031	35	■ 0.988	e \circ x=x	$e \circ x = x$	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .								
kohlhase-omdoc_F00032	35	■ 0.988	x \in G	$x \in G$	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .								
kohlhase-omdoc_F00033	35	■ 0.988	e	e	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .								
kohlhase-omdoc_F00034	35	■ 0.988	S	S	$e \circ x = x$ for all $x \in G$. e is called a left unit of S .								
kohlhase-omdoc_F00035	35	■ 0.915	f	f	Proof: We assume that there is another left unit $f \dots$								
kohlhase-omdoc_F00036	35	—	\square	$\square$	—								
kohlhase-omdoc_F00037	37	■ 0.491	\mathrm{BCF}^+	BCF^+	MIZAR [Rud92] and Ω MEGA [BCF ⁺ 97] which utilize strong logical systems								
kohlhase-omdoc_F00038	38	■ 0.995	\left[\mathrm{SBB}^+02\right]	$\left[\mathrm{SBB}^+02\right]$	on the other. Ω MEGA [SBB ⁺ 02] is a large-scale proof development environ-								
kohlhase-omdoc_F00039	40	■ 0.772	1{ }^{\text{st}} 2000	1 st 2000	Version 1.0 of the OMDoc format was released on November 1 st 2000 to								
kohlhase-omdoc_F00040	40	■ 0.981	29{ }^{\text{th}} 2001	29 th 2001	OMDoc 1.1 was released on December 29 th 2001 as an attempt to roll								
kohlhase-omdoc_F00041 [conf 0.000]	41	■ 0.000	X+0=0</ \mathrm{FMP}>	$X + 0 = 0 < / \mathrm{FMP} >$	<table><tr><th>Level of Representation</th><th>OMDoc Example</th></tr><tr><td><i>Theory Level:</i> Development Graph – Inheritance via symbol-mapping – Theory inclusion via proof-obligations – Local (one-step) vs. global links </td><td></td></tr><tr><td><i>Statement Level:</i> – Axiom, definition, theorem, proof, example,... – Structure explicit in statement forms and references </td><td><definition for="#plus" type="recursive"> <CMP>Addition is defined by recursion on the second argument </CMP> <FMP>X + 0 = 0</FMP> <FMP>X + s(Y) = s(X + Y)</FMP> </definition></td></tr><tr><td><i>Object Level:</i> OPENMATH/MATHML – Objects as logical formulae – Semantics by pointing to theory level </td><td><OMA> <OMS cd="arith1" name="plus" /> <OMV name="X" /> <OMS cd="nat" name="zero" /> </OMA></td></tr></table>	Level of Representation	OMDoc Example	<i>Theory Level:</i> Development Graph – Inheritance via symbol-mapping – Theory inclusion via proof-obligations – Local (one-step) vs. global links		<i>Statement Level:</i> – Axiom, definition, theorem, proof, example,... – Structure explicit in statement forms and references	<definition for="#plus" type="recursive"> <CMP>Addition is defined by recursion on the second argument </CMP> <FMP>X + 0 = 0</FMP> <FMP>X + s(Y) = s(X + Y)</FMP> </definition>	<i>Object Level:</i> OPENMATH/MATHML – Objects as logical formulae – Semantics by pointing to theory level	<OMA> <OMS cd="arith1" name="plus" /> <OMV name="X" /> <OMS cd="nat" name="zero" /> </OMA>
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<i>Object Level:</i> OPENMATH/MATHML – Objects as logical formulae – Semantics by pointing to theory level	<OMA> <OMS cd="arith1" name="plus" /> <OMV name="X" /> <OMS cd="nat" name="zero" /> </OMA>												

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kohlhase-omdoc_F00064	49	■0.997	$x^{\{y\}}$	x^y	under these laws being denoted respectively by $x + y$, xy , or $x.y$ and x^y) (<i>Set</i>																					
kohlhase-omdoc_F00065	49	■1.000	$(X, Y) \mapsto X \circ Y$	$(X, Y) \mapsto X \circ Y$	(3) Let E be a set; the mapping $(X, Y) \mapsto X \circ Y$ is a law of composition																					
kohlhase-omdoc_F00066	49	■1.000	$(f, g) \mapsto f \circ g$	$(f, g) \mapsto f \circ g$	mapping $(f, g) \mapsto f \circ g$ is a law of composition on the set of mappings from E																					
kohlhase-omdoc_F00067	52	■0.221	$\mathrm{APT}_{\mathrm{E}} \mathrm{X}$	$\mathrm{APT}_{\mathrm{E}} \mathrm{X}$	<table><tr><td>24,25</td><td>The <code>cc:permissions</code> element gives the world the permission to reproduce and distribute it freely. Furthermore the license grants the public the right to make derivative works under certain conditions.</td><td>12.3 p. 103</td></tr><tr><td>26</td><td>The <code>cc:prohibitions</code> can be used to prohibit certain uses of the document, but this one is unencumbered.</td><td>12.3 p. 103</td></tr><tr><td>27</td><td>The <code>cc:requirements</code> states conditions under which the license is granted. In our case the licensee is required to keep the copyright notice and license notices intact during distribution, to give credit to the copyright holder, and that any derivative works derived from this document must be licensed under the same terms as this document (the copyleft clause).</td><td>12.3 p. 103</td></tr><tr><td>31-37</td><td>The <code>omtext</code> element is used to mark up text fragments. Here, we have simply used a single <code>omtext</code> to classify the whole text in the fragment as unspecific “text”.</td><td>14.3 p. 124</td></tr><tr><td>32-36</td><td>The <code>CMP</code> element holds the actual text in a multilingual group. Its <code>xml:lang</code> specifies the language. If the document is used with a DTD or an XML schema (as we are) this attribute is redundant, since the default value given by the DTD or schema is <code>en</code>. More keywords in other languages can be given by adding more <code>CMP</code> elements.</td><td>14.1 p. 122</td></tr><tr><td>33–35</td><td>The text of the $\mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ fragment we are migrating. For simplicity we do not change the text, and leave that to later stages of the migration.</td><td></td></tr><tr><td>38</td><td>The closing tag of the root <code>omdoc</code> element. There may not be text after this in the file.</td><td>11.1 p. 90</td></tr></table>	24,25	The <code>cc:permissions</code> element gives the world the permission to reproduce and distribute it freely. Furthermore the license grants the public the right to make derivative works under certain conditions.	12.3 p. 103	26	The <code>cc:prohibitions</code> can be used to prohibit certain uses of the document, but this one is unencumbered.	12.3 p. 103	27	The <code>cc:requirements</code> states conditions under which the license is granted. In our case the licensee is required to keep the copyright notice and license notices intact during distribution, to give credit to the copyright holder, and that any derivative works derived from this document must be licensed under the same terms as this document (the copyleft clause).	12.3 p. 103	31-37	The <code>omtext</code> element is used to mark up text fragments. Here, we have simply used a single <code>omtext</code> to classify the whole text in the fragment as unspecific “text”.	14.3 p. 124	32-36	The <code>CMP</code> element holds the actual text in a multilingual group. Its <code>xml:lang</code> specifies the language. If the document is used with a DTD or an XML schema (as we are) this attribute is redundant, since the default value given by the DTD or schema is <code>en</code> . More keywords in other languages can be given by adding more <code>CMP</code> elements.	14.1 p. 122	33–35	The text of the $\mathrm{L}^{\mathrm{A}}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ fragment we are migrating. For simplicity we do not change the text, and leave that to later stages of the migration.		38	The closing tag of the root <code>omdoc</code> element. There may not be text after this in the file.	11.1 p. 90
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kohlhase-omdoc_F00068	59	■0.999	λ	λ	a λ -term $\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$. The value <code>simple</code> of the <code>type</code> at-																					
kohlhase-omdoc_F00069	59	■0.999	$\lambda E, F. \operatorname{set}(E) \wedge F : E \times E \rightarrow E^2$	$\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$	a λ -term $\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E^2$. The value <code>simple</code> of the <code>type</code> at-																					
kohlhase-omdoc_F00070	59	■1.000	(A, B)	(A, B)	it occurs. So if we have <code>law_of_composition(A, B)</code> somewhere this can be																					
kohlhase-omdoc_F00071	59	■0.863	$(\lambda E, F. \operatorname{set}(E) \wedge F : E \times E \rightarrow E)(A, B)$	$(\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E)(A, B)$	reduced to $(\lambda E, F. \mathit{set}(E) \wedge F : E \times E \rightarrow E)(A, B)$ which in turn reduces ³																					
kohlhase-omdoc_F00072	59	■0.917	$\operatorname{set}(A) \wedge B : A \times A \rightarrow A$	$\mathit{set}(A) \wedge B : A \times A \rightarrow A$	to $\mathit{set}(A) \wedge B : A \times A \rightarrow A$ or in other words <code>law_of_composition(A, B)</code> is																					
kohlhase-omdoc_F00073	59	■1.000	A	A	true, iff A is a set and B is a function from $A \times A$ to A . This definition is																					
kohlhase-omdoc_F00074	59	■1.000	B	B	true, iff A is a set and B is a function from $A \times A$ to A . This definition is																					
kohlhase-omdoc_F00075	59	■1.000	$A \times A$	$A \times A$	true, iff A is a set and B is a function from $A \times A$ to A . This definition is																					
kohlhase-omdoc_F00076	59	■0.819	$\lambda X. A$	$\lambda X. A$	the form $\lambda X. A$ to an argument B , then the result is obtained by binding all the																					
kohlhase-omdoc_F00077	59	■1.000	X	X	formal parameters X to the actual parameter B , i.e. the result is the value of A ,																					
kohlhase-omdoc_F00078	60	■0.967	M	M	in other words: M is a magma, iff it is a pair (E, F) , where F is a law of																					
kohlhase-omdoc_F00079	60	■0.967	(E, F)	(E, F)	in other words: M is a magma, iff it is a pair (E, F) , where F is a law of																					
kohlhase-omdoc_F00080	60	■0.967	F	F	in other words: M is a magma, iff it is a pair (E, F) , where F is a law of																					
kohlhase-omdoc_F00081	62	■0.986	$\left[\mathrm{BCC}^{+04} \right]$	$\left[\mathrm{BCC}^{+04} \right]$	dard $\left[\mathrm{BCC}^{+04} \right]$ to codify knowledge about mathematical symbols and con-																					

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kohlhase-omdoc_F00082	66	■ 1.000	n	n	The symbol to represent the n -ary function to return the least common mul-
kohlhase-omdoc_F00083	66	■ 0.981	$\operatorname{lcm}(a, b)$	$\operatorname{lcm}(a, b)$	We define $\operatorname{lcm}(a, b)$ as $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$
kohlhase-omdoc_F00084	66	■ 0.981	$\frac{a \cdot b}{\operatorname{gcd}(a, b)}$	$\frac{a \cdot b}{\operatorname{gcd}(a, b)}$	We define $\operatorname{lcm}(a, b)$ as $\frac{a \cdot b}{\operatorname{gcd}(a, b)}$
kohlhase-omdoc_F00085	66	■ 0.940	a	a	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_F00086	66	■ 0.940	b	b	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_F00087	66	■ 0.940	$c > 0$	$c > 0$	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_F00088	66	■ 0.940	$(a \mid c)$	$(a \mid c)$	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_F00089	66	■ 0.940	$(b \mid c)$	$(b \mid c)$	Lemma 3. For all integers a, b there is no $c > 0$ such that $(a c)$ and $(b c)$ and
kohlhase-omdoc_F00090	66	■ 1.000	$c < \operatorname{lcm}(a, b)$	$c < \operatorname{lcm}(a, b)$	$c < \operatorname{lcm}(a, b)$.
kohlhase-omdoc_F00091	66	■ 0.998	$\operatorname{kgV}(a, b)$	$\operatorname{kgV}(a, b)$	Wir definieren $\operatorname{kgV}(a, b)$ als $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$
kohlhase-omdoc_F00092	66	■ 0.998	$\frac{a \cdot b}{\operatorname{ggT}(a, b)}$	$\frac{a \cdot b}{\operatorname{ggT}(a, b)}$	Wir definieren $\operatorname{kgV}(a, b)$ als $\frac{a \cdot b}{\operatorname{ggT}(a, b)}$
kohlhase-omdoc_F00093	66	■ 0.354	$b \mid c$	$b \mid c$	Lemma 3. Für alle ganzen Zahlen a, b gibt es kein $c > 0$ mit $(a c)$ und $(b c)$
kohlhase-omdoc_F00094	66	■ 0.531	$c < \operatorname{kgV}(a, b)$	$c < \operatorname{kgV}(a, b)$	und $c < \operatorname{kgV}(a, b)$.
kohlhase-omdoc_F00095	68	■ 1.000	$\mathbb{N}$	$\mathbb{N}$	more elaborate definition of $\mathbb{N}$). The sortdef element posits that the set of
kohlhase-omdoc_F00096	68	■ 1.000	$s(s(s(0)))$	$s(s(s(0)))$	instance the number three is represented as $s(s(s(0)))$, where s stands for
kohlhase-omdoc_F00097	68	■ 1.000	s	s	instance the number three is represented as $s(s(s(0)))$, where s stands for
kohlhase-omdoc_F00098	68	■ 1.000	$\leq$	$\leq$	Finally we have extended the natural numbers by an ordering function $\leq$
kohlhase-omdoc_F00099	69	■ 0.887	TO^1	TO^1	theory, as witnessed by the assertion $\operatorname{leq}.\mathrm{TO}^1$.
kohlhase-omdoc_F00100	72	■ 1.000	$\sigma(\circ)$	$\sigma(\circ)$	together with the converse $\sigma(\circ)$ of $\circ$ is also a semigroup: the obligation for
kohlhase-omdoc_F00101	72	■ 1.000	$\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$	$\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$	$\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$, which is logically equivalent
kohlhase-omdoc_F00102	72	■ 0.835	τx	τx	scription operator τx .: If there is a unique x , such that A is true, then the
kohlhase-omdoc_F00103	73	■ 1.000	ρ	ρ	ρ in Figure 7.1), in OMDOC we can share this structure using the base on
kohlhase-omdoc_F00104	73	■ 0.951	$\cdot^{-1}$	$\cdot^{-1}$	function $\cdot^{-1}$ from that via a definite description.
kohlhase-omdoc_F00105	74	■ 1.000	σ	σ	represented by σ and τ in Figure 7.1). These rename all of the imported
kohlhase-omdoc_F00106	74	■ 1.000	τ	τ	represented by σ and τ in Figure 7.1). These rename all of the imported
kohlhase-omdoc_F00107	74	■ 0.301	$\cdot^*$	$\cdot^*$	the punctuated set constructor $\cdot^*$, which is imported from the additive group
kohlhase-omdoc_F00108	74	■ 0.267	l	l	$\langle \text{morphism} \rangle M \mapsto R, x \circ y \mapsto x * y, e \mapsto 1, \cdot^{-1} \mapsto - \langle / \text{morphism} \rangle$
kohlhase-omdoc_F00109	84	■ 1.000	$\mathcal{N}$	$\mathcal{N}$	imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00110	84	■ 1.000	$\mathcal{F}$	$\mathcal{F}$	imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this
kohlhase-omdoc_F00111	84	■ 1.000	$\mathcal{E}_{\mathcal{N}}$	$\mathcal{E}_{\mathcal{N}}$	imation $\mathcal{N}$ of the real theory $\mathcal{F}$, only to show an example $\mathcal{E}_{\mathcal{N}}$ of where this
kohlhase-omdoc_F00112	84	■ 1.000	$\mathcal{S}$	$\mathcal{S}$	is insufficient. Then we propose a first (straw-man) solution $\mathcal{S}$, and show an
kohlhase-omdoc_F00113	84	■ 1.000	$\mathcal{E}_{\mathcal{S}}$	$\mathcal{E}_{\mathcal{S}}$	example $\mathcal{E}_{\mathcal{S}}$ of why this does not work. Based on the information we gleaned
kohlhase-omdoc_F00114	84	■ 1.000	$\mathcal{E}_{\mathcal{F}}$	$\mathcal{E}_{\mathcal{F}}$	theory and demonstrate that this works on $\mathcal{E}_{\mathcal{F}}$.
kohlhase-omdoc_F00115	84	■ 1.000	$\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}, \mathcal{E}_{\mathcal{F}}$	$\mathcal{N}, \mathcal{E}_{\mathcal{N}}, \mathcal{S}, \mathcal{E}_{\mathcal{S}}, \mathcal{F}$	sents the content structure, where the boxes $\mathcal{N}$, $\mathcal{E}_{\mathcal{N}}$, $\mathcal{S}$, $\mathcal{E}_{\mathcal{S}}$, $\mathcal{F}$, and $\mathcal{E}_{\mathcal{F}}$ signify
kohlhase-omdoc_F00116	84	■ 0.384	$\operatorname{slides} \mathrm{sl}_i$	slides sl_i	the slides sl_i are grouped into a lecture. The dashed lines between the two
kohlhase-omdoc_F00117	84	■ 1.000	n_1	n_1	example: the text fragment n_1 links the content $\mathcal{E}_{\mathcal{N}}$, which is referenced from
kohlhase-omdoc_F00118	84	■ 0.974	n_2	n_2	the content structure, to slide 1. The fragment n_2 might say something like
kohlhase-omdoc_F00119	85	■ 0.986	$\left[\operatorname{MBG}^{+03}\right]$	$\left[\operatorname{MBG}^{+03}\right]$	the ACTIVEMATH system $\left[\operatorname{MBG}^{+03}\right]$ generates a simple narrative structure
kohlhase-omdoc_F00120	87	■ 0.810	9.3^2	9.3^2	in Listing 9.3 ² to the URL which uniquely identifies the service object in the
kohlhase-omdoc_F00121	103	■ 1.000	R	R	Let R be a ref element of type include . We call the element the URI
kohlhase-omdoc_F00122	103	■ 0.978	$\left[\operatorname{SBC}^{+00}\right]$	$\left[\operatorname{SBC}^{+00}\right]$	system $\left[\operatorname{SBC}^{+00}\right]$ can — for instance — extract document fragments based on
kohlhase-omdoc_F00123	107	■ 0.910	$\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle \mathrm{T} \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$	$\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle \mathrm{T} \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$	Concretely, the format is $\langle YYY Y \rangle - \langle MM \rangle - \langle DD \rangle \mathrm{T} \langle hh \rangle : \langle mm \rangle : \langle ss \rangle$
kohlhase-omdoc_F00124	107	■ 0.781	$\langle YYY Y \rangle$	$\langle YYY Y \rangle$	where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the
kohlhase-omdoc_F00125	107	■ 0.781	$\langle MM \rangle$	$\langle MM \rangle$	where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the
kohlhase-omdoc_F00126	107	■ 0.781	$\langle DD \rangle$	$\langle DD \rangle$	where $\langle YYY Y \rangle$ represents the year, $\langle MM \rangle$ the month, and $\langle DD \rangle$ the
kohlhase-omdoc_F00127	107	■ 0.582	$\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$	$\langle hh \rangle, \langle mm \rangle, \langle ss \rangle$	date/time separator and $\langle hh \rangle$, $\langle mm \rangle$, $\langle ss \rangle$ represent hour, minutes, and
kohlhase-omdoc_F00128	107	■ 0.260	$\langle ss \rangle . \langle sss \rangle$	$\langle ss \rangle . \langle sss \rangle$	cision of fractional seconds if desired, i.e the format $\langle ss \rangle . \langle sss \rangle$ with
kohlhase-omdoc_F00129	108	■ 1.000	$\widehat{=}$	$\widehat{=}$	be an ISO 639 norm two-letter language specifier, like en $\widehat{=}$ English,
kohlhase-omdoc_F00130	108	■ 0.327	$\mathrm{de} \widehat{=}$	$\mathrm{de} \widehat{=}$	de $\widehat{=}$ German, fr $\widehat{=}$ French, nl $\widehat{=}$ Dutch,
kohlhase-omdoc_F00131	114	■ 0.586	$^{+03}$	$^{+03}$	and the MATHML 2.0 recommendation (second edition) $\left[\operatorname{ABC}^{+03}\right]$ are part
kohlhase-omdoc_F00132	116	■ 1.000	$\sin(x)$	$\sin(x)$	For instance, the formula $\sin(x)$ would be modeled as an application of
kohlhase-omdoc_F00133	116	■ 1.000	$\forall x. \sin(x) \leq \pi$	$\forall x. \sin(x) \leq \pi$	formula $\forall x. \sin(x) \leq \pi$.
kohlhase-omdoc_F00134	117	■ 0.999	$x + \pi$	$x + \pi$	In the OPENMATH fragment in Listing 13.3 the expression $x + \pi$ is annotated
kohlhase-omdoc_F00135	118	■ 1.000	$\sin(\pi)$	$\sin(\pi)$	faulty representation of $\sin(\pi)$. The outer error flags an arity violation (the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00136	118	■ 1.000	π	π	the representation of the constant π (we used the name po instead of pi).
kohlhase-omdoc_F00137	121	■ 0.973	14^{th}	14^{th}	metics (Kindergarten to 14^{th} grade (i.e. undergraduate college level)).
kohlhase-omdoc_F00138	121	■ 0.630	$2.0\left[\mathrm{ABC}^{+}03\right]$	$2.0\left[\mathrm{ABC}^{+}03\right]$	second edition of MATHML 2.0 [ABC ⁺ 03], a maintenance release for the
kohlhase-omdoc_F00139	121	■ 0.997	$\left[\mathrm{ABC}^{+}03\right]$	$\left[\mathrm{ABC}^{+}03\right]$	MATHML 2.0 that are relevant to OMDoc; for the full story see [ABC ⁺ 03].
kohlhase-omdoc_F00140	121	■ 0.456	$^{\{7\}}$	7	The top-level element of MATHML is the m:math ⁷ element, see Figure 13.10
kohlhase-omdoc_F00141	124	■ 1.000	$(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$	$(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$	e.g. the type $(\mathbb{R} \rightarrow \mathbb{R}) \rightarrow (\mathbb{R} \rightarrow \mathbb{R})$ for functionals on the real numbers. In
kohlhase-omdoc_F00142	126	■ 0.397	$\mathrm{cd} =$	$\mathrm{cd} =$	tent dictionary identified in the cd attribute. A symbol $\langle \mathrm{cd} = \langle cd_1 \rangle$
kohlhase-omdoc_F00143	126	■ 0.397	$\left\langle \left\langle cd_1 \right\rangle \right\rangle$	$\langle \langle cd_1 \rangle \rangle$	tent dictionary identified in the cd attribute. A symbol $\langle \mathrm{cd} = \langle cd_1 \rangle$
kohlhase-omdoc_F00144	126	■ 0.175	$\left\langle \left\langle cd_2 \right\rangle \right\rangle$	$\langle \langle cd_2 \rangle \rangle$	$\mathrm{name} = \langle \langle name_1 \rangle \rangle />$ is equal to $\langle \mathrm{cd} = \langle cd_2 \rangle \rangle$ $\mathrm{name} = \langle \langle name_2 \rangle \rangle />$, iff
kohlhase-omdoc_F00145	126	■ 0.175	$\left\langle \left\langle name_2 \right\rangle \right\rangle$	$\langle \langle name_2 \rangle \rangle$	$\mathrm{name} = \langle \langle name_1 \rangle \rangle />$ is equal to $\langle \mathrm{cd} = \langle cd_2 \rangle \rangle$ $\mathrm{name} = \langle \langle name_2 \rangle \rangle />$, iff
kohlhase-omdoc_F00146	126	■ 0.973	$\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$	$\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$	$\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$ and $\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$ as XML simple names. In case of vari-
kohlhase-omdoc_F00147	126	■ 0.973	$\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$	$\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$	$\langle \langle cd_1 \rangle \rangle = \langle \langle cd_2 \rangle \rangle$ and $\langle \langle name_1 \rangle \rangle = \langle \langle name_2 \rangle \rangle$ as XML simple names. In case of vari-
kohlhase-omdoc_F00148	126	■ 0.923	α	α	named without changing the object (α-conversion). If $\langle \mathrm{OMV} \ \mathrm{name} = \mathrm{x} \rangle />$
kohlhase-omdoc_F00149	129	■ 0.990	CMP^2	CMP^2	is represented as a groups of CMP ² elements which contain the vernacular
kohlhase-omdoc_F00150	130	■ 0.942	FMP^4	FMP^4	An FMP ⁴ element is the general element for representing formal mathematical
kohlhase-omdoc_F00151	131	■ 0.287	$V, F. \mathrm{Im}(F) = V$	$V, F. \mathrm{Im}(F) = V$	$\langle \mathrm{FMP} \ \mathrm{logic} = \mathrm{hol} \rangle \binop = \lambda V, F. \mathrm{Im}(F) = V \wedge \mathrm{Dom}(F) = V \langle / \mathrm{FMP} \rangle$
kohlhase-omdoc_F00152	131	■ 0.404	$A \langle / \mathrm{FMP} \rangle$	$A \langle / \mathrm{FMP} \rangle$	is equivalent to $\langle \mathrm{FMP} \rangle A \langle / \mathrm{FMP} \rangle$, where A is an OPENMATH, Content-MATHML,
kohlhase-omdoc_F00153	134	■ 0.562	$0 \in \mathbb{N}$	$0 \in \mathbb{N}$	ken ⁵ . Let us consider an example: We can equivalently say “ $0 \in \mathbb{N}$ ” and “the
kohlhase-omdoc_F00154	137	■ 0.995	$\underline{\mathrm{ul}}$	$\underline{\mathrm{ul}}$	<i>Unordered Lists</i> ul is the constructor for unordered or bulleted lists, the in
kohlhase-omdoc_F00155	140	■ 1.000	$s(n)$	$s(n)$	2. The successor $s(n)$ of a natural number n is a natural number.
kohlhase-omdoc_F00156	140	■ 0.997	$0, s$	$0, s$	structures that interpret 0, s , and $\mathbb{N}$ and satisfy these axioms must be isomor-
kohlhase-omdoc_F00157	141	■ 1.000	$\mathcal{T}$	$\mathcal{T}$	of a theory $\mathcal{T}$ by just looking at its form: for instance, axioms of the form $s =$
kohlhase-omdoc_F00158	141	■ 1.000	$s =$	$s =$	of a theory $\mathcal{T}$ by just looking at its form: for instance, axioms of the form $s =$
kohlhase-omdoc_F00159	141	■ 0.623	$\mathbf{A}$	A	A , where s is a symbol that does not occur in $\mathcal{T}$ and A is a formula containing
kohlhase-omdoc_F00160	141	■ 1.000	$s = \mathbf{A}$	$s = \mathbf{A}$	symbols. The axiom $s = \mathbf{A}$ only constrains the meaning of the new symbol
kohlhase-omdoc_F00161	141	■ 1.000	$\mathcal{A}$	$\mathcal{A}$	Generally an axiom $\mathcal{A}$ that results in a conservative extension is called

Identifier	Page	Conf.	LaTeX source	Rendered	Image												
kohlhase-omdoc_F00162	141	0.554	$1 := s(0)$	$1 := s(0)$	<table><tr><th>Definiendum</th><th>Definiens</th><th>Type</th></tr><tr><td>The number 1</td><td>$1 := s(0)$ (1 is the successor of 0)</td><td>simple</td></tr><tr><td>The exponential function e'</td><td>The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function +</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table>	Definiendum	Definiens	Type	The number 1	$1 := s(0)$ (1 is the successor of 0)	simple	The exponential function e'	The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive
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The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive															
kohlhase-omdoc_F00163	141	0.554	$e^{\cdot}$	e'	<table><tr><th>Definiendum</th><th>Definiens</th><th>Type</th></tr><tr><td>The number 1</td><td>$1 := s(0)$ (1 is the successor of 0)</td><td>simple</td></tr><tr><td>The exponential function e'</td><td>The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function +</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table>	Definiendum	Definiens	Type	The number 1	$1 := s(0)$ (1 is the successor of 0)	simple	The exponential function e'	The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive
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kohlhase-omdoc_F00164	141	0.554	$\partial f = f[$	$\partial f = f[$	<table><tr><th>Definiendum</th><th>Definiens</th><th>Type</th></tr><tr><td>The number 1</td><td>$1 := s(0)$ (1 is the successor of 0)</td><td>simple</td></tr><tr><td>The exponential function e'</td><td>The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function +</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table>	Definiendum	Definiens	Type	The number 1	$1 := s(0)$ (1 is the successor of 0)	simple	The exponential function e'	The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive
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The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive															
kohlhase-omdoc_F00165	141	0.554	$f(0)=1]$	$f(0) = 1]$	<table><tr><th>Definiendum</th><th>Definiens</th><th>Type</th></tr><tr><td>The number 1</td><td>$1 := s(0)$ (1 is the successor of 0)</td><td>simple</td></tr><tr><td>The exponential function e'</td><td>The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function +</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table>	Definiendum	Definiens	Type	The number 1	$1 := s(0)$ (1 is the successor of 0)	simple	The exponential function e'	The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive
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kohlhase-omdoc_F00166	141	0.554	$x+0=x$	$x + 0 = x$	<table><tr><th>Definiendum</th><th>Definiens</th><th>Type</th></tr><tr><td>The number 1</td><td>$1 := s(0)$ (1 is the successor of 0)</td><td>simple</td></tr><tr><td>The exponential function e'</td><td>The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function +</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table>	Definiendum	Definiens	Type	The number 1	$1 := s(0)$ (1 is the successor of 0)	simple	The exponential function e'	The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive
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The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive															
kohlhase-omdoc_F00167	141	0.554	$x+s(y)=s(x+y)$	$x + s(y) = s(x + y)$	<table><tr><th>Definiendum</th><th>Definiens</th><th>Type</th></tr><tr><td>The number 1</td><td>$1 := s(0)$ (1 is the successor of 0)</td><td>simple</td></tr><tr><td>The exponential function e'</td><td>The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].</td><td>implicit</td></tr><tr><td>The addition function +</td><td>Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.</td><td>recursive</td></tr></table>	Definiendum	Definiens	Type	The number 1	$1 := s(0)$ (1 is the successor of 0)	simple	The exponential function e'	The exponential function e' is the solution to the differential equation $\partial f = f$ [where $f(0) = 1$].	implicit	The addition function +	Addition on the natural numbers is defined by the equations $x + 0 = x$ and $x + s(y) = s(x + y)$.	recursive
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kohlhase-omdoc_F00168	143	0.993	$1.1\backslash\mathrm{left[\mathrm{BPSM}^{\scriptscriptstyle+} 04\mathrm{right}]}$	$1.1 \left[\text{BPSM}^+04 \right]$	1.1 [BPSM ⁺ 04]. The optional attribute scope takes the values global and												
kohlhase-omdoc_F00169	144	1.000	$e: t$	$e : t$	processes. A type declaration $e:t$ makes the information that a symbol or												
kohlhase-omdoc_F00170	145	0.998	t	t	expression e is in a set represented by a type t available to a specified math-												
kohlhase-omdoc_F00171	145	1.000	$\sum_{i=1}^n a_{\{i\}} x^{\{i\}}$	$\sum_{i=1}^n a_i x^i$	or that expressions of the form $\sum_{i=1}^n a_i x^i$ are polynomials, if the a_i are real												
kohlhase-omdoc_F00172	145	1.000	$a_{\{i\}}$	a_i	or that expressions of the form $\sum_{i=1}^n a_i x^i$ are polynomials, if the a_i are real												
kohlhase-omdoc_F00173	145	0.999	$\forall\mathrm{all} \ x_{\{1\}}, \ \backslash\mathrm{ldots}, \ x_{\{n\}} \ . \ e \ \backslash\mathrm{in} \ t$	$\forall x_1, \dots, x_n. e \in t$	$\forall x_1, \dots, x_n. e \in t$ for a type declaration $e:t$ with variables $x_1, \dots, x_n$. If the												

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kohlhase-omdoc_ F00174	145	0.999	$x_{\{1\}}, \ldots, x_{\{n\}}$	$x_1, \dots, x_n$	$\forall x_1, \dots, x_n. e \in t$ for a type declaration $e:t$ with variables $x_1, \dots, x_n$. If the																																			
kohlhase-omdoc_ F00175	148	0.397	$\rangle\rangle^*$	$\rangle\rangle^*$	<table><tr><th>Element</th><th colspan="2">Attributes</th><th>D</th><th>Content</th></tr><tr><td></td><th>Required</th><th>Optional</th><th>C</th><td></td></tr><tr><td>assertion</td><td></td><td>xml:id, for, type, theory, class, style, status, just-by</td><td>+</td><td>CMP*, FMP*</td></tr><tr><td>type</td><td>system</td><td>xml:id, for, just-by, theory, class, style</td><td>−</td><td>CMP*, $\langle\langle mobj \rangle\rangle, \langle\langle mobj \rangle\rangle$</td></tr><tr><td>example</td><td>for</td><td>xml:id, type, assertion, theory, class, style</td><td>+</td><td>CMP* $\langle\langle mobj \rangle\rangle^*$</td></tr><tr><td>alternative</td><td>for, theory, entailed-by, entails, entailed-by-thm, entails-thm</td><td>xml:id, type, theory, class, style</td><td>+</td><td>CMP*, (FMP* reequation* $\langle\langle mobj \rangle\rangle$)</td></tr><tr><td colspan="5">where $\langle\langle mobj \rangle\rangle$ is (OMOBJ m:math legacy)</td></tr></table>	Element	Attributes		D	Content		Required	Optional	C		assertion		xml:id, for, type, theory, class, style, status, just-by	+	CMP*, FMP*	type	system	xml:id, for, just-by, theory, class, style	−	CMP*, $\langle\langle mobj \rangle\rangle, \langle\langle mobj \rangle\rangle$	example	for	xml:id, type, assertion, theory, class, style	+	CMP* $\langle\langle mobj \rangle\rangle^*$	alternative	for, theory, entailed-by, entails, entailed-by-thm, entails-thm	xml:id, type, theory, class, style	+	CMP*, (FMP* reequation* $\langle\langle mobj \rangle\rangle$)	where $\langle\langle mobj \rangle\rangle$ is (OMOBJ m:math legacy)				
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example	for	xml:id, type, assertion, theory, class, style	+	CMP* $\langle\langle mobj \rangle\rangle^*$																																				
alternative	for, theory, entailed-by, entails, entailed-by-thm, entails-thm	xml:id, type, theory, class, style	+	CMP*, (FMP* reequation* $\langle\langle mobj \rangle\rangle$)																																				
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kohlhase-omdoc_F00193	152	■0.997	$\mathcal{E}$	$\mathcal{E}$	objects in OMDoc. Conceptually, we model an example $\mathcal{E}$ as a pair $(\mathcal{W}, \mathbf{A})$,																										
kohlhase-omdoc_F00194	152	■0.997	$(\mathcal{W}, \mathbf{A})$	$(\mathcal{W}, \mathbf{A})$	objects in OMDoc. Conceptually, we model an example $\mathcal{E}$ as a pair $(\mathcal{W}, \mathbf{A})$,																										
kohlhase-omdoc_F00195	152	■0.992	$\mathcal{W} = \left(\mathcal{W}_{\{1\}}, \ldots, \mathcal{W}_{\{n\}} \right)$	$\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$	where $\mathcal{W} = (\mathcal{W}_1, \dots, \mathcal{W}_n)$ is an n -tuple of mathematical objects and $\mathbf{A}$ is an																										
kohlhase-omdoc_F00196	152	■0.923	$\mathbf{S}$	$\mathbf{S}$	symbol $\mathbf{S}$, then $\mathbf{A}$ must be of the form $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$.																										
kohlhase-omdoc_F00197	152	■0.923	$\mathbf{S} \left(\mathcal{W}_{\{1\}}, \ldots, \mathcal{W}_{\{n\}} \right)$	$\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$	symbol $\mathbf{S}$, then $\mathbf{A}$ must be of the form $\mathbf{S}(\mathcal{W}_1, \dots, \mathcal{W}_n)$.																										
kohlhase-omdoc_F00198	152	■0.999	$\mathbf{C}$	$\mathbf{C}$	tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$,																										
kohlhase-omdoc_F00199	152	■0.999	$\mathcal{Q} \ \mathbf{D}$	$\mathcal{Q}\mathbf{D}$	tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$,																										
kohlhase-omdoc_F00200	152	■0.999	$\mathbf{D}$	$\mathbf{D}$	tion more carefully. We assume that $\mathbf{C}$ is of the form $\mathcal{Q}\mathbf{D}$ for some formula $\mathbf{D}$,																										
kohlhase-omdoc_F00201	152	■1.000	$\mathcal{Q}$	$\mathcal{Q}$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1 W_1, \dots, \mathcal{Q}_m W_m$ of $m \geq n = \#\mathcal{W}$ quantifications of																										
kohlhase-omdoc_F00202	152	■1.000	$\mathcal{Q}_{\{1\}} W_{\{1\}}, \ldots, \mathcal{Q}_{\{m\}} W_{\{m\}}$	$\mathcal{Q}_1 W_1, \dots, \mathcal{Q}_m W_m$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1 W_1, \dots, \mathcal{Q}_m W_m$ of $m \geq n = \#\mathcal{W}$ quantifications of																										
kohlhase-omdoc_F00203	152	■1.000	$m \geq n = \# \ \mathcal{W}$	$m \geq n = \#\mathcal{W}$	where $\mathcal{Q}$ is a sequence $\mathcal{Q}_1 W_1, \dots, \mathcal{Q}_m W_m$ of $m \geq n = \#\mathcal{W}$ quantifications of																										
kohlhase-omdoc_F00204	152	■0.999	$\mathcal{Q}_{\{i\}}$	$\mathcal{Q}_i$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-																										
kohlhase-omdoc_F00205	152	■0.999	$\forall$	$\forall$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-																										
kohlhase-omdoc_F00206	152	■0.999	$\exists$	$\exists$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image																														
kohlhase-omdoc_F00207	152	■ 0.999	$\mathcal{Q}^{\prime}$	$\mathcal{Q}'$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-																														
kohlhase-omdoc_F00208	152	■ 0.999	$m - n$	$m - n$	using quantifiers $\mathcal{Q}_i$ like $\forall$ or $\exists$. Now let $\mathcal{Q}'$ be a sub-sequence of $m - n$ quan-																														
kohlhase-omdoc_F00209	152	■ 0.985	$\mathbf{D}^{\prime}$	$\mathbf{D}'$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the W_{i_j} such that the $\mathcal{Q}_{i_j}$ are absent																														
kohlhase-omdoc_F00210	152	■ 0.985	W_{i_j}	W_{i_j}	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the W_{i_j} such that the $\mathcal{Q}_{i_j}$ are absent																														
kohlhase-omdoc_F00211	152	■ 0.985	$\mathcal{Q}_{i_j}$	$\mathcal{Q}_{i_j}$	tifiers of $\mathcal{Q}$ and $\mathbf{D}'$ be $\mathbf{D}$ only that all the W_{i_j} such that the $\mathcal{Q}_{i_j}$ are absent																														
kohlhase-omdoc_F00212	152	■ 0.998	$\mathcal{W}_j$	$\mathcal{W}_j$	from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$. If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$,																														
kohlhase-omdoc_F00213	152	■ 0.998	$1 \leq j \leq n$	$1 \leq j \leq n$	from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$. If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$,																														
kohlhase-omdoc_F00214	152	■ 0.998	$\mathcal{E} = (\mathcal{W}, \mathbf{A})$	$\mathcal{E} = (\mathcal{W}, \mathbf{A})$	from $\mathcal{Q}'$ have been replaced by $\mathcal{W}_j$ for $1 \leq j \leq n$. If $\mathcal{E} = (\mathcal{W}, \mathbf{A})$ supports $\mathbf{C}$,																														
kohlhase-omdoc_F00215	152	■ 1.000	$\mathbf{A} = \mathcal{Q}'\mathbf{D}'$	$\mathbf{A} = \mathcal{Q}'\mathbf{D}'$	then $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$ and if $\mathcal{E}$ is a counter-example for $\mathbf{C}$, then $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$.																														
kohlhase-omdoc_F00216	152	■ 1.000	$\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$	$\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$	then $\mathbf{A} = \mathcal{Q}'\mathbf{D}'$ and if $\mathcal{E}$ is a counter-example for $\mathbf{C}$, then $\mathbf{A} = \neg\mathcal{Q}'\mathbf{D}'$.																														
kohlhase-omdoc_F00217	154	■ 1.000	A^*	A^*	alphabet A as an argument and returns the set A^* of stringss over A , concat																														
kohlhase-omdoc_F00218	154	■ 0.914	ϵ	ϵ	the empty string ϵ . Then we state that $\mathcal{W} = (A^*, ::, \epsilon)$ is a monoid in an																														
kohlhase-omdoc_F00219	154	■ 0.914	$\mathcal{W} = (A^*, ::, \epsilon)$	$\mathcal{W} = (A^*, ::, \epsilon)$	the empty string ϵ . Then we state that $\mathcal{W} = (A^*, ::, \epsilon)$ is a monoid in an																														
kohlhase-omdoc_F00220	154	■ 1.000	$(\mathcal{W}, \mathbf{A}')$	$(\mathcal{W}, \mathbf{A}')$	mon.ex2 uses the pair $(\mathcal{W}, \mathbf{A}')$ as a counter-example to the false conjecture																														
kohlhase-omdoc_F00221	154	■ 0.380	$M,$	$M,$	<CMP>In a monoid M , the left unit and the right unit coincide,</CMP>																														
kohlhase-omdoc_F00222	156	■ 0.939	$(\langle\langle top + thc \rangle\rangle) *$	$(\langle\langle top + thc \rangle\rangle) *$	<table><tr><th>Element</th><th colspan="2">Attributes</th><th>D</th><th>Content</th></tr><tr><td></td><th>Req.</th><th>Optional</th><td>C</td><td></td></tr><tr><td>theory</td><td></td><td>xml:id, class, style, cdbase, cdversion, cdrevision, cdstatus, cdurl, cdreviewdate</td><td>+</td><td>$(\langle\langle top + thc \rangle\rangle \mid imports) *$</td></tr><tr><td>imports</td><td>from</td><td>id, type, class, style</td><td>+</td><td></td></tr><tr><td>tgroup</td><td></td><td>xml:id, modules, type, class, style</td><td>+</td><td>$(\langle\langle top + thc \rangle\rangle) *$</td></tr><tr><td colspan="5">where $\langle\langle top + thc \rangle\rangle$ stands for top-level and theory-constitutive elements</td></tr></table>	Element	Attributes		D	Content		Req.	Optional	C		theory		xml:id, class, style, cdbase, cdversion, cdrevision, cdstatus, cdurl, cdreviewdate	+	$(\langle\langle top + thc \rangle\rangle \mid imports) *$	imports	from	id, type, class, style	+		tgroup		xml:id, modules, type, class, style	+	$(\langle\langle top + thc \rangle\rangle) *$	where $\langle\langle top + thc \rangle\rangle$ stands for top-level and theory-constitutive elements				
Element	Attributes		D	Content																															
	Req.	Optional	C																																
theory		xml:id, class, style, cdbase, cdversion, cdrevision, cdstatus, cdurl, cdreviewdate	+	$(\langle\langle top + thc \rangle\rangle \mid imports) *$																															
imports	from	id, type, class, style	+																																
tgroup		xml:id, modules, type, class, style	+	$(\langle\langle top + thc \rangle\rangle) *$																															
where $\langle\langle top + thc \rangle\rangle$ stands for top-level and theory-constitutive elements																																			
kohlhase-omdoc_F00223	161	■ 1.000	$\emptyset$	$\emptyset$	tors $\emptyset$ and the set insertion function ι , since the set $\{a\}$ can be obtained																														
kohlhase-omdoc_F00224	161	■ 1.000	ι	ι	tors $\emptyset$ and the set insertion function ι , since the set $\{a\}$ can be obtained																														
kohlhase-omdoc_F00225	161	■ 1.000	$\{a\}$	$\{a\}$	tors $\emptyset$ and the set insertion function ι , since the set $\{a\}$ can be obtained																														
kohlhase-omdoc_F00226	161	■ 1.000	$\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$	$\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$	e.g. $\iota(a, \emptyset) = \iota(a, \iota(a, \emptyset))$. This kind of sort is called generated , since it only																														
kohlhase-omdoc_F00227	162	■ 1.000	$\mathbb{P}$	$\mathbb{P}$	encoding. In this example, we introduce a second sort $\mathbb{P}$ for positive nat-																														
kohlhase-omdoc_F00228	166	■ 1.000	P	P	Proof: We need to prove that the set P of all prime numbers is not																														
kohlhase-omdoc_F00229	166	■ 1.000	$P = \{p_1, \dots, p_n\}$	$P = \{p_1, \dots, p_n\}$	3. Then $P = \{p_1, \dots, p_n\}$ for some p_i .																														
kohlhase-omdoc_F00230	166	■ 1.000	p_i	p_i	3. Then $P = \{p_1, \dots, p_n\}$ for some p_i .																														
kohlhase-omdoc_F00231	166	■ 1.000	$q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$	$q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$	4. Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$.																														
kohlhase-omdoc_F00232	166	■ 1.000	$p_i \in P$	$p_i \in P$	5. Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.																														

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_F00233	166	■ 1.000	$q > p_{\{i\}}$	$q > p_i$	5. Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.
kohlhase-omdoc_F00234	166	■ 1.000	$q \notin P$	$q \notin P$	5. Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.
kohlhase-omdoc_F00235	166	■ 1.000	q	q	6. We prove the absurdity by showing that q is prime:
kohlhase-omdoc_F00236	166	■ 1.000	$q = p_{\{i\}} k + 1$	$q = p_i k + 1$	7. For each $p_i \in P$ we have $q = p_i k + 1$ for some natural
kohlhase-omdoc_F00237	166	■ 1.000	k	k	number k , so p_i can not divide q ;
kohlhase-omdoc_F00238	166	■ 0.990	$\bar{\lambda}\mu\tilde{\mu}$	$\bar{\lambda}\mu\tilde{\mu}$	by interpreting it as an implementation of the $\bar{\lambda}\mu\tilde{\mu}$ calculus [Coe05] proof
kohlhase-omdoc_F00239	168	■ 1.000	$12\ f\ f$	$12ff$	the proof. See for example lines $12ff$ in Listing 17.2 for examples of
kohlhase-omdoc_F00240	170	■ 0.986	$\left[\mathrm{BCF}^{+97}\right]$	$\left[\mathrm{BCF}^{+97}\right]$	multiple levels of abstraction). This proof representation (see [BCF ⁺ 97] for a
kohlhase-omdoc_F00241	179	■ 0.791	$(R, +)$	$(R, +)$	R together with two operations $+$ and $*$, such that $(R, +)$ is a group with
kohlhase-omdoc_F00242	179	■ 1.000	$\left(R^{*}, *\right)$	$(R^{*}, *)$	unit 0 and inverse operation $-$ and $(R^{*}, *)$ is a monoid with unit 1 and base
kohlhase-omdoc_F00243	179	■ 1.000	$R^{*}:=\{r \in R \mid r \neq 0\}$	$R^{*} := \{r \in R \mid r \neq 0\}$	set $R^{*}:=\{r \in R \mid r \neq 0\}$. Using the literal inheritance regime introduced so
kohlhase-omdoc_F00244	179	■ 0.604	$M=(S, \circ, e)$	$M = (S, \circ, e)$	² On any monoid $M = (S, \circ, e)$, we have the $\cdot^{*}$ operator, which converts a set
kohlhase-omdoc_F00245	179	■ 1.000	$S \subseteq M$	$S \subseteq M$	$S \subseteq M$ in to $S^{*}:=\{r \in S \mid r \neq e\}$
kohlhase-omdoc_F00246	179	■ 1.000	$S^{*}:=\{r \in S \mid r \neq e\}$	$S^{*} := \{r \in S \mid r \neq e\}$	$S \subseteq M$ in to $S^{*}:=\{r \in S \mid r \neq e\}$
kohlhase-omdoc_F00247	180	■ 1.000	$\{ \}^{\{3\}} t_{\{1\}}, \ldots, t_{\{n\}}$	$^3 t_1, \dots, t_n$	Let $\mathcal{S}$ be a theory with theory-constitutive elements ³ $t_1, \dots, t_n$ and $\sigma: \mathcal{S} \rightarrow \mathcal{T}$
kohlhase-omdoc_F00248	180	■ 1.000	$\sigma: \mathcal{S} \rightarrow \mathcal{T}$	$\sigma: \mathcal{S} \rightarrow \mathcal{T}$	Let $\mathcal{S}$ be a theory with theory-constitutive elements ³ $t_1, \dots, t_n$ and $\sigma: \mathcal{S} \rightarrow \mathcal{T}$
kohlhase-omdoc_F00249	180	■ 1.000	$\sigma(t_i)$	$\sigma(t_i)$	constitutive statements $\sigma(t_i)$ from $\mathcal{S}$. For instance, the theory of rings inherits
kohlhase-omdoc_F00250	180	■ 1.000	$\forall x. x + 0 = x$	$\forall x. x + 0 = x$	the axiom $\forall x. x + 0 = x$ from the theory of monoids as $\sigma(\forall x. x + 0 = x) =$
kohlhase-omdoc_F00251	180	■ 1.000	$\sigma(\forall x. x + 0 = x) =$	$\sigma(\forall x. x + 0 = x) =$	the axiom $\forall x. x + 0 = x$ from the theory of monoids as $\sigma(\forall x. x + 0 = x) =$
kohlhase-omdoc_F00252	180	■ 0.999	$\forall x. x * 1 = x$	$\forall x. x * 1 = x$	$\forall x. x * 1 = x$.
kohlhase-omdoc_F00253	181	■ 0.999	$R, +, 0, -, *, 1$	$R, +, 0, -, *, 1$	the morphism. In our example, the symbols $R, +, 0, -, *, 1$ are visible in the
kohlhase-omdoc_F00254	181	■ 1.000	L	L	concept of a sorting function, i.e. a function that — given a list L as input
kohlhase-omdoc_F00255	181	■ 1.000	$L^{\{\prime\}}$	L'	— returns a returns a permutation L' of L that is ordered. In the situation
kohlhase-omdoc_F00256	182	■ 0.879	σ_i	σ_i	σ_i with <code>xml:id="⟨refi⟩"</code> , such that the target theory of σ_{i-1} is the source
kohlhase-omdoc_F00257	182	■ 0.879	σ_{i-1}	σ_{i-1}	σ_i with <code>xml:id="⟨refi⟩"</code> , such that the target theory of σ_{i-1} is the source
kohlhase-omdoc_F00258	182	■ 0.902	$\mathcal{S}, \sigma(\mathbf{S})$	$\mathcal{S}, \sigma(\mathbf{S})$	constitutive statement $\mathbf{S}$ of $\mathcal{S}$, $\sigma(\mathbf{S})$ is provable in $\mathcal{T}$ (we say that $\sigma(\mathbf{S})$ is a
kohlhase-omdoc_F00259	182	■ 0.902	$\sigma(\mathbf{S})$	$\sigma(\mathbf{S})$	constitutive statement $\mathbf{S}$ of $\mathcal{S}$, $\sigma(\mathbf{S})$ is provable in $\mathcal{T}$ (we say that $\sigma(\mathbf{S})$ is a
kohlhase-omdoc_F00260	182	■ 0.905	$\mathbf{T}$	$\mathbf{T}$	in theory $\mathcal{T}$ via σ , if there is an assertional statement $\mathbf{T}$ in $\mathcal{T}$, such that the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00261	182	■ 1.000	$\mathcal{T}^5$	$\mathcal{T}^5$	discussed in Section 18.1 induces a theory inclusion from $\mathcal{S}$ into $\mathcal{T}^5$ (the
kohlhase-omdoc_ F00262	183	■ 1.000	$\sigma(\mathbf{A})$	$\sigma(\mathbf{A})$	$\mathcal{S}$ -theorem for for all $\mathcal{T}$ -theorems of the from $\sigma(\mathbf{A})$. If the morphism σ is the
kohlhase-omdoc_ F00263	183	■ 0.421	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	$\langle \text{assertion xml:id="conv.assoc"} \forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x \rangle$
kohlhase-omdoc_ F00264	183	■ 0.421	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	$\forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x$	$\langle \text{assertion xml:id="conv.assoc"} \forall x, y, z \in M. z \circ (y \circ x) = (z \circ y) \circ x \rangle$
kohlhase-omdoc_ F00265	183	■ 0.765	$\forall x, y \in M. y \circ x \in M$	$\forall x, y \in M. y \circ x \in M$	$\langle \text{assertion xml:id="conv.closed"} \text{ theory="semigroup"} \forall x, y \in M. y \circ x \in M \rangle$
kohlhase-omdoc_ F00266	183	■ 0.325	$\forall x \in M. e \circ x = x$	$\forall x \in M. e \circ x = x$	$\langle \text{assertion xml:id="left.unit"} \text{ theory="monoid"} \forall x \in M. e \circ x = x \rangle$
kohlhase-omdoc_ F00267	186	■ 0.998	$\left[\mathrm{BCD}^{+02} \right]$	$\left[\mathrm{BCD}^{+02} \right]$	(see e.g. [BCD ⁺ 02] for an overview over the branch cut situation). Reading
kohlhase-omdoc_ F00268	187	■ 1.000	Γ	Γ	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00269	187	■ 1.000	$\mathcal{T}_1$	$\mathcal{T}_1$	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00270	187	■ 1.000	$\mathcal{T}_2$	$\mathcal{T}_2$	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00271	187	■ 1.000	$\mathcal{C}_1$	$\mathcal{C}_1$	in the top theory inclusion Γ . On the basis of theories $\mathcal{T}_1$ and $\mathcal{T}_2$, theory $\mathcal{C}_1$
kohlhase-omdoc_ F00272	187	■ 0.999	$\mathcal{A}_1$	$\mathcal{A}_1$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00273	187	■ 0.999	$\mathcal{B}_1$	$\mathcal{B}_1$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00274	187	■ 0.999	$\mathcal{C}_2$	$\mathcal{C}_2$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00275	187	■ 0.999	$\mathcal{A}_2$	$\mathcal{A}_2$	is built up via theories $\mathcal{A}_1$ and $\mathcal{B}_1$. Similarly, theory $\mathcal{C}_2$ is built up via $\mathcal{A}_2$
kohlhase-omdoc_ F00276	187	■ 1.000	$\mathcal{B}_2$	$\mathcal{B}_2$	and $\mathcal{B}_2$ (in the latter, we have a non-trivial non-trivial morphism σ). Let us
kohlhase-omdoc_ F00277	187	■ 1.000	$\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$	$\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_ F00278	187	■ 1.000	$\mathcal{X}_1$	$\mathcal{X}_1$	assume for the sake of this argument that for $\mathcal{X}_i \in \{\mathcal{A}, \mathcal{B}, \mathcal{C}\}$ theories $\mathcal{X}_1$ and
kohlhase-omdoc_ F00279	187	■ 1.000	$\mathcal{X}_2$	$\mathcal{X}_2$	$\mathcal{X}_2$ are so similar that axiom inclusions (they are indicated by thin dashed
kohlhase-omdoc_ F00280	187	■ 0.995	α, β	α, β	arrows in Figure 18.8 and have the formula-mappings α, β , and γ) are easy
kohlhase-omdoc_ F00281	187	■ 0.995	γ	γ	arrows in Figure 18.8 and have the formula-mappings α, β , and γ) are easy
kohlhase-omdoc_ F00282	187	■ 1.000	$\beta(\mathbf{B})$	$\beta(\mathbf{B})$	constitutive for $\mathcal{C}_1$, say that it is local in $\mathcal{B}_1$, then we already know that $\beta(\mathbf{B})$ is
kohlhase-omdoc_ F00283	187	■ 1.000	β	β	provable in $\mathcal{B}_2$ since β is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$
kohlhase-omdoc_ F00284	187	■ 1.000	$\sigma(\beta(\mathbf{B}))$	$\sigma(\beta(\mathbf{B}))$	provable in $\mathcal{B}_2$ since β is an axiom inclusion. Moreover, we know that $\sigma(\beta(\mathbf{B}))$
kohlhase-omdoc_ F00285	187	■ 1.000	$\Gamma = \sigma \circ \beta$	$\Gamma = \sigma \circ \beta$	have $\Gamma = \sigma \circ \beta$, then we are done for $\mathbf{B}$ and in fact for all local statements
kohlhase-omdoc_ F00286	187	■ 1.000	Γ^{-1}	Γ^{-1}	(and often also in the converse theory inclusion Γ^{-1} that would prove that the
kohlhase-omdoc_ F00287	188	■ 1.000	$\tau := \sigma_n \circ \dots \circ \sigma_1 \circ \sigma$	$\tau := \sigma_n \circ \dots \circ \sigma_1 \circ \sigma$	morphism $\tau := \sigma_n \circ \dots \circ \sigma_1 \circ \sigma$), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local
kohlhase-omdoc_ F00288	188	■ 1.000	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$	morphism $\tau := \sigma_n \circ \dots \circ \sigma_1 \circ \sigma$), if $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}_1$ is an axiom inclusion or (local

Identifier	Page	Conf.	LaTeX source	Rendered	Image																					
kohlhase-omdoc_F00289	188	1.000	$\mathcal{T}_{\{i\}} \xrightarrow{\sigma_i} \mathcal{T}_{\{i+1\}}$	$\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}$	theory import) and $\mathcal{T}_i \xrightarrow{\sigma_i} \mathcal{T}_{i+1}$ are theory inclusions (or local theory import).																					
kohlhase-omdoc_F00290	188	1.000	$\mathcal{T}_{\{1\}}, \mathcal{T}_{\{2\}}, \mathcal{A}_{\{1\}}$	$\mathcal{T}_1, \mathcal{T}_2, \mathcal{A}_1$	justify four new axiom inclusions from the theories $\mathcal{T}_1$, $\mathcal{T}_2$, $\mathcal{A}_1$, and $\mathcal{B}_1$ into																					
kohlhase-omdoc_F00291	188	1.000	$\mathcal{X}$	$\mathcal{X}$	Thus, for each theory $\mathcal{X}$ that $\mathcal{C}_1$ inherits from, there is an axiom inclusion																					
kohlhase-omdoc_F00292	188	1.000	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$	$\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$	We call a situation where we have a formula mapping $\mathcal{S} \xrightarrow{\sigma} \mathcal{T}$, and																					
kohlhase-omdoc_F00293	188	1.000	$\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$	$\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$	an axiom inclusion $\mathcal{X} \xrightarrow{\sigma_{\mathcal{X}}} \mathcal{T}$ for every theory $\mathcal{X}$ that $\mathcal{S}$ inherits from a																					
kohlhase-omdoc_F00294	188	1.000	$\sigma_{\mathcal{X}}$	$\sigma_{\mathcal{X}}$	decomposition for σ , if the $\sigma_{\mathcal{X}}$ and σ are compatible. As we have seen in																					
kohlhase-omdoc_F00295	190	0.996	i	<i>i</i>	there must be theory inclusions σ_i with <code>xml:id="refi"</code> , such that the target																					
kohlhase-omdoc_F00296	190	1.000	σ_1	σ_1	theory of σ_1 is the source theory of σ_2 .																					
kohlhase-omdoc_F00297	190	1.000	σ_2	σ_2	theory of σ_1 is the source theory of σ_2 .																					
kohlhase-omdoc_F00298	197	1.000	$\forall P, Q. P \vee Q \Rightarrow Q \vee P$	$\forall P, Q. P \vee Q \Rightarrow Q \vee P$	$\forall P, Q. P \vee Q \Rightarrow Q \vee P$ assuming appropriate templates for implication and																					
kohlhase-omdoc_F00299	199	1.000	g c d	<i>gcd</i>	<i>gcd</i> in English but <i>ggT</i> (“größter gemeinsamer Teiler”) in German). Obvi-																					
kohlhase-omdoc_F00300	199	1.000	g g T	<i>ggT</i>	<i>gcd</i> in English but <i>ggT</i> (“größter gemeinsamer Teiler”) in German). Obvi-																					
kohlhase-omdoc_F00301	200	1.000	$f^{\{\prime\}}$	<i>f'</i>	<i>f'</i> . The case <code>infix</code> is reserved for binary operators, where the function is																					
kohlhase-omdoc_F00302	200	1.000	$(x+2)^{\{2\}}$	$(x+2)^2$	are presented as $(x+2)^2$ and $x+y^2$, respectively.																					
kohlhase-omdoc_F00303	200	1.000	$x+y^{\{2\}}$	$x+y^2$	are presented as $(x+2)^2$ and $x+y^2$, respectively.																					
kohlhase-omdoc_F00304	201	0.952	$*, \wedge, \cap$	$*, \wedge, \cap$	<table><tr><th>Precedence</th><th>Operators</th><th>Comment</th></tr><tr><td>200</td><td>$+, -$</td><td>unary</td></tr><tr><td>200</td><td>$^$</td><td>exponentiation</td></tr><tr><td>400</td><td>$*, \wedge, \cap$</td><td>multiplicative</td></tr><tr><td>500</td><td>$+, -, \vee, \cup$</td><td>additive</td></tr><tr><td>600</td><td>$/$</td><td>fraction</td></tr><tr><td>700</td><td>$=, \neq, \leq, <, >, \geq$</td><td>relation</td></tr></table>	Precedence	Operators	Comment	200	$+, -$	unary	200	$^$	exponentiation	400	$*, \wedge, \cap$	multiplicative	500	$+, -, \vee, \cup$	additive	600	$/$	fraction	700	$=, \neq, \leq, <, >, \geq$	relation
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kohlhase-omdoc_F00305	201	0.952	$=, \neq, \leq, <, >, \geq$	$=, \neq, \leq, <, >, \geq$	<table><tr><th>Precedence</th><th>Operators</th><th>Comment</th></tr><tr><td>200</td><td>$+, -$</td><td>unary</td></tr><tr><td>200</td><td>$^$</td><td>exponentiation</td></tr><tr><td>400</td><td>$*, \wedge, \cap$</td><td>multiplicative</td></tr><tr><td>500</td><td>$+, -, \vee, \cup$</td><td>additive</td></tr><tr><td>600</td><td>$/$</td><td>fraction</td></tr><tr><td>700</td><td>$=, \neq, \leq, <, >, \geq$</td><td>relation</td></tr></table>	Precedence	Operators	Comment	200	$+, -$	unary	200	$^$	exponentiation	400	$*, \wedge, \cap$	multiplicative	500	$+, -, \vee, \cup$	additive	600	$/$	fraction	700	$=, \neq, \leq, <, >, \geq$	relation
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kohlhase-omdoc_F00306	201	1.000	(a, b)	(a, b)	presentation. For instance, in the notation (a, b) for pairs, the binary																					
kohlhase-omdoc_F00307	201	0.912	$\mathrm{IAT}_{\{\mathrm{E}\}} \mathrm{X}$	$\mathrm{IAT}_{\mathrm{E}}\mathrm{X}$	of the figure: the $\mathrm{IAT}_{\mathrm{E}}\mathrm{X}$ and the HTML presentations have attached																					

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																																																					
					<table><thead><tr><th>Notation specification</th><th>Example</th></tr></thead><tbody><tr><td><pre><presentation for="#forall" role="binding" separator="." > <use format="TeX">\forallall</use> <use format="html">&#8704;</use> </presentation></pre></td><td><pre><OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMV name="X" /> </OMBVAR> <OMS cd="logic1" name="true" /> </OMBIND></pre></td></tr><tr><td colspan="5"></td><td> Using XSLT templates induced from the presentation element on the OPEN-MATH expression yields $\forall X.\text{true}$, where the glyph $\forall$ carries a hyperlink⁶ to its definition, as the crossref-symbol on the presentation element has the default value yes. Internally, the hyperlinks are format-dependent, we have: $\text{\LaTeX}$: <code>\href{../ocd/logic1.ps#true}{\forallall}X.</code> <code>\href{../ocd/logic1.ps#true}{\sf true}</code> HTML: <code>&#8704; X.</code> <code>true</code> </td></tr></tbody></table> <table><tr><td>kohlhase-omdoc_ F00308 [conf 0.000]</td><td>202</td><td>0.000</td><td><code>\forallall X</code></td><td>$\forall X$</td><td></td></tr><tr><td>kohlhase-omdoc_ F00309 [conf 0.000]</td><td>202</td><td>0.000</td><td><code>{ }^{\varnothing}</code></td><td>$\varnothing$</td><td></td></tr><tr><td>kohlhase-omdoc_ F00310</td><td>202</td><td>1.000</td><td><code>\binom{n}{m}</code></td><td>$\binom{n}{m}$</td><td>For instance, the binomial coefficient is some presented as $\binom{n}{m}$ (spoken</td></tr><tr><td>kohlhase-omdoc_ F00311</td><td>202</td><td>1.000</td><td><code>m</code></td><td>m</td><td>“n choose m”) and represented as</td></tr><tr><td>kohlhase-omdoc_ F00312</td><td>204</td><td>0.995</td><td><code>X_{4}</code></td><td>X_4</td><td>tifier would be represented as X_4, yielding $\forall X_4.X_4 = X_4$ which is exactly</td></tr><tr><td>kohlhase-omdoc_ F00313</td><td>204</td><td>0.995</td><td><code>\forallall X_{4} . X_{4}=X_{4}</code></td><td>$\forall X_4.X_4 = X_4$</td><td>tifier would be represented as X_4, yielding $\forall X_4.X_4 = X_4$ which is exactly</td></tr><tr><td>kohlhase-omdoc_ F00314</td><td>205</td><td>0.982</td><td><code>\forallall \prec, X . X \prec X</code></td><td>$\forall \prec, X.X \prec X$</td><td>This would give us the presentation $\forall \prec, X.X \prec X$. Here, the first occurrence</td></tr><tr><td>kohlhase-omdoc_ F00315</td><td>205</td><td>1.000</td><td><code>\prec</code></td><td>$\prec$</td><td>of the variable $\prec$ is handled by the second notation declaration (it does not</td></tr><tr><td>kohlhase-omdoc_ F00316</td><td>215</td><td>0.755</td><td><code>\mathrm{T}_{\mathrm{E}} \mathrm{XBook}</code></td><td>$\text{T}_{\text{E}}\text{XBook}$</td><td>Listing 21.2. An exercise from the $\text{T}_{\text{E}}\text{XBook}$</td></tr><tr><td>kohlhase-omdoc_ F00317</td><td>215</td><td>0.891</td><td><code>{\, ,p\mid\hbox{</code></td><td><i>(not rendered)</i></td><td>$\{ \, ,p\mid\hbox{\\$p\\$ and \\$p+2\\$ are prime}\, , \}$</td></tr><tr><td>kohlhase-omdoc_ F00318</td><td>215</td><td>0.891</td><td><code>and</code></td><td><i>and</i></td><td>$\{ \, ,p\mid\hbox{\\$p\\$ and \\$p+2\\$ are prime}\, , \}$</td></tr><tr><td>kohlhase-omdoc_ F00319</td><td>215</td><td>0.891</td><td><code>are prime}\, , }</code></td><td><i>(not rendered)</i></td><td>$\{ \, ,p\mid\hbox{\\$p\\$ and \\$p+2\\$ are prime}\, , \}$</td></tr><tr><td>kohlhase-omdoc_ F00320</td><td>215</td><td>0.468</td><td><code>\{ \, ,p\mid p</code></td><td>$\{ p \mid p$</td><td>$\{ \, ,p\mid p^{\sim}$ and $\\$p+2\\$ are prime\, , \}$</td></tr><tr><td>kohlhase-omdoc_ F00321</td><td>215</td><td>0.468</td><td><code>p+2</code></td><td>$p + 2$</td><td>$\{ \, ,p\mid p^{\sim}$ and $\\$p+2\\$ are prime\, , \}$</td></tr><tr><td>kohlhase-omdoc_ F00322</td><td>215</td><td>0.468</td><td><code>\, , }</code></td><td></td><td>$\{ \, ,p\mid p^{\sim}$ and $\\$p+2\\$ are prime\, , \}$</td></tr><tr><td>kohlhase-omdoc_ F00323</td><td>215</td><td>0.405</td><td><code>\{ \, ,p\mid p\{\rm and\}p+2\rm\ are\ prime\, , }</code></td><td><i>(not rendered)</i></td><td>$\\$ \{ \, ,p\mid p\{ \rm and\} \ p+2\rm\ are\ prime\, , \} \\$</phrase>$</td></tr></table>	Notation specification	Example	<pre><presentation for="#forall" role="binding" separator="." > <use format="TeX">\forallall</use> <use format="html">&#8704;</use> </presentation></pre>	<pre><OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMV name="X" /> </OMBVAR> <OMS cd="logic1" name="true" /> </OMBIND></pre>						Using XSLT templates induced from the presentation element on the OPEN-MATH expression yields $\forall X.\text{true}$, where the glyph $\forall$ carries a hyperlink⁶ to its definition, as the crossref-symbol on the presentation element has the default value yes. 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X_{4}=X_{4}</code>	$\forall X_4.X_4 = X_4$	tifier would be represented as X_4 , yielding $\forall X_4.X_4 = X_4$ which is exactly	kohlhase-omdoc_ F00314	205	0.982	<code>\forallall \prec, X . X \prec X</code>	$\forall \prec, X.X \prec X$	This would give us the presentation $\forall \prec, X.X \prec X$. Here, the first occurrence	kohlhase-omdoc_ F00315	205	1.000	<code>\prec</code>	$\prec$	of the variable $\prec$ is handled by the second notation declaration (it does not	kohlhase-omdoc_ F00316	215	0.755	<code>\mathrm{T}_{\mathrm{E}} \mathrm{XBook}</code>	$\text{T}_{\text{E}}\text{XBook}$	Listing 21.2. An exercise from the $\text{T}_{\text{E}}\text{XBook}$	kohlhase-omdoc_ F00317	215	0.891	<code>{\, ,p\mid\hbox{</code>	<i>(not rendered)</i>	$\{ \, ,p\mid\hbox{\$p\$ and \$p+2\$ are prime}\, , \}$	kohlhase-omdoc_ F00318	215	0.891	<code>and</code>	<i>and</i>	$\{ \, ,p\mid\hbox{\$p\$ and \$p+2\$ are prime}\, , \}$	kohlhase-omdoc_ F00319	215	0.891	<code>are prime}\, , }</code>	<i>(not rendered)</i>	$\{ \, ,p\mid\hbox{\$p\$ and \$p+2\$ are prime}\, , \}$	kohlhase-omdoc_ F00320	215	0.468	<code>\{ \, ,p\mid p</code>	$\{ p \mid p$	$\{ \, ,p\mid p^{\sim}$ and $\$p+2\$ are prime\, , \}$	kohlhase-omdoc_ F00321	215	0.468	<code>p+2</code>	$p + 2$	$\{ \, ,p\mid p^{\sim}$ and $\$p+2\$ are prime\, , \}$	kohlhase-omdoc_ F00322	215	0.468	<code>\, , }</code>		$\{ \, ,p\mid p^{\sim}$ and $\$p+2\$ are prime\, , \}$	kohlhase-omdoc_ F00323
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kohlhase-omdoc_ F00319	215	0.891	<code>are prime}\, , }</code>	<i>(not rendered)</i>	$\{ \, ,p\mid\hbox{\$p\$ and \$p+2\$ are prime}\, , \}$																																																																																																					
kohlhase-omdoc_ F00320	215	0.468	<code>\{ \, ,p\mid p</code>	$\{ p \mid p$	$\{ \, ,p\mid p^{\sim}$ and $\$p+2\$ are prime\, , \}$																																																																																																					
kohlhase-omdoc_ F00321	215	0.468	<code>p+2</code>	$p + 2$	$\{ \, ,p\mid p^{\sim}$ and $\$p+2\$ are prime\, , \}$																																																																																																					
kohlhase-omdoc_ F00322	215	0.468	<code>\, , }</code>		$\{ \, ,p\mid p^{\sim}$ and $\$p+2\$ are prime\, , \}$																																																																																																					
kohlhase-omdoc_ F00323	215	0.405	<code>\{ \, ,p\mid p\{\rm and\}p+2\rm\ are\ prime\, , }</code>	<i>(not rendered)</i>	$\$ \{ \, ,p\mid p\{ \rm and\} \ p+2\rm\ are\ prime\, , \} \$</phrase>$																																																																																																					

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kohlhase-omdoc_ F00324	218	■ 1.000	$\mathcal{K}$	$\mathcal{K}$	For a formal definition let $\mathcal{K}$ be a set of documents. We take a docu-
kohlhase-omdoc_ F00325	218	■ 0.935	$\{ \}^{\{1\}} \mathcal{K}$	${}^1\mathcal{K}$	ment model to be a partial equivalence relation ¹ $\mathcal{K}$ on documents, such
kohlhase-omdoc_ F00326	218	■ 0.998	$\{d \mid d \mathcal{K} d\} = \mathcal{K}$	$\{d \mid d\mathcal{K}d\} = \mathcal{K}$	that $\{d \mid d\mathcal{K}d\} = \mathcal{K}$. In particular, a relation $\mathcal{K}$ is an equivalence relation on
kohlhase-omdoc_ F00327	218	■ 1.000	d	d	$\mathcal{K}$. For a given document model $\mathcal{K}$, let us say that two documents d and d'
kohlhase-omdoc_ F00328	218	■ 1.000	$d^{\{\prime\}}$	d'	$\mathcal{K}$. For a given document model $\mathcal{K}$, let us say that two documents d and d'
kohlhase-omdoc_ F00329	218	■ 1.000	$d \mathcal{K} d^{\{\prime\}}$	$d\mathcal{K}d'$	are $\mathcal{K}$-equal , iff $d\mathcal{K}d'$. We call a property p $\mathcal{K}$-invariant , iff for all $d\mathcal{K}d'$, p
kohlhase-omdoc_ F00330	218	■ 1.000	$p \mathcal{K} X$	$p\mathcal{K}$	are $\mathcal{K}$-equal , iff $d\mathcal{K}d'$. We call a property p $\mathcal{K}$-invariant , iff for all $d\mathcal{K}d'$, p
kohlhase-omdoc_ F00331	218	■ 1.000	$d \mathcal{K} X d^{\{\prime\}}$, p	$d\mathcal{K}d', p$	are $\mathcal{K}$-equal , iff $d\mathcal{K}d'$. We call a property p $\mathcal{K}$-invariant , iff for all $d\mathcal{K}d'$, p
kohlhase-omdoc_ F00332	218	■ 1.000	p	p	holds on d whenever p holds on d' .
kohlhase-omdoc_ F00333	218	■ 1.000	$[d]_{\mathcal{K}}$	$[d]_{\mathcal{K}}$	¹ A partial equivalence relation is a symmetric transitive relation. We will use $[d]_{\mathcal{K}}$
kohlhase-omdoc_ F00334	218	■ 0.999	$[d]_{\mathcal{K}} := \{e \mid d \mathcal{K} e\}$	$[d]_{\mathcal{K}} := \{e \mid d\mathcal{K}e\}$	for the equivalence class of d , i.e. $[d]_{\mathcal{K}} := \{e \mid d\mathcal{K}e\}$
kohlhase-omdoc_ F00335	220	■ 1.000	$X+Y=$	$X + Y =$	<i>not</i> use mathematical equality here, which would make the formula $X + Y =$
kohlhase-omdoc_ F00336	220	■ 0.268	$Y+X$	$Y + X$	$Y + X$ (the om:OMOBJ with xml:id="comm1" in Listing 22.3 instantiated with
kohlhase-omdoc_ F00337	220	■ 1.000	$X+Y=X+Y$	$X + Y = X + Y$	addition for op) mathematically equal to the trivial condition $X+Y = X+Y$,
kohlhase-omdoc_ F00338	220	■ 1.000	$X+Y$	$X + Y$	obtained by exchanging the right hand side $Y + X$ of the equality by $X + Y$,
kohlhase-omdoc_ F00339	232	■ 0.999	$\left[\mathrm{ABD}^{+01}\right]$	$\left[\mathrm{ABD}^{+01}\right]$	tions for DTD modularization $[\mathrm{ABD}^{+01}]$. This partitions the DTD into DTD
kohlhase-omdoc_ F00340	232	■ 0.966	$\{ \}^{+01}$	$^{+01}$	process, we will briefly describe the module structure. Following $[\mathrm{ABD}^{+01}]$,
kohlhase-omdoc_ F00341	234	■ 0.988	$d\rangle$	$d\rangle$	as omdoc- $\langle\!\!\langle SIId \!\!\rangle\!\rangle$.rnc, where $\langle\!\!\langle SIId \!\!\rangle\!\rangle$ stands for the sub-language identifier.
kohlhase-omdoc_ F00342	238	■ 1.000	$2\ n$	$2n$	so achieve interoperability for n systems with only $2n$ translations instead of
kohlhase-omdoc_ F00343	238	■ 1.000	$n^{\{2\}}$	n^2	n^2 . As we have seen in Section 2.1, OPENMATH and Content-MATHML pro-
kohlhase-omdoc_ F00344	240	■ 0.970	$\lambda C l a m$	$\lambda C lam$	systems Ω_{MEGA} [BCF ⁺ 97], INKA [HS96], Pvs [ORS92], $\lambda C lam$ [RSG98b],
kohlhase-omdoc_ F00345	240	■ 0.836	$\left[\mathrm{ABI}^{+96}\right]$	$\left[\mathrm{ABI}^{+96}\right]$	TPS [ABI ⁺ 96] and CoQ [Tea] see Section 26.18
kohlhase-omdoc_ F00346	241	■ 0.183	$\mathrm{IAT}_{\{E\}} \mathrm{X}$	$\mathrm{IAT}_E\mathrm{X}$	omdoc2html.xsl for XHTML+MATHML and omdoc2tex.xsl for $\mathbb{L}\mathrm{T}_{\mathrm{E}}\mathrm{X}$.
kohlhase-omdoc_ F00347	243	■ 0.990	$\mathrm{T}_{\{\mathrm{E}\}} \mathrm{X}_{\{\text{MACS}\}}$	$\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\text{MACS}}$	plugin for MOZILLA-based browsers, and the plugin for $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\text{MACS}}$ (Sec-
kohlhase-omdoc_ F00348	243	■ 1.000	$\mathrm{ST}_{\{\mathrm{E}\}} \mathrm{X}$	$\mathrm{ST}_{\mathrm{E}}\mathrm{X}$	OMDOC that can be generated by the qmath parser. $\mathbb{S}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ defines a
kohlhase-omdoc_ F00349	243	■ 0.220	$_{\{\text{E}\}}$	E	XHTML+MATHML or $\mathbb{L}\mathrm{T}_{\mathrm{E}}\mathrm{X}$, which can in turn be published on the
kohlhase-omdoc_ F00350	246	■ 1.000	$2+2=4$	$2 + 2 = 4$	Freedom is the freedom to say $\$2+2=4\$$.
kohlhase-omdoc_ F00351	251	■ 0.429	$\$ \mathrm{e}^{\{\wedge\}(\mathrm{pi} * \mathrm{i})+1=0} \$$	$\$e^{(\mathrm{pi} * \mathrm{i}) + 1 = 0}\$$	is similar to typing $\mathrm{T}_{\mathrm{E}}\mathrm{X}$ or QMATH text: one can type $\$e^{(\mathrm{pi} * \mathrm{i})+1=0}\$$ and

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kohlhase-omdoc_F00352	251	■ 1.000	$e^{\pi i}+1=0$	$e^{\pi i}+1=0$	get $e^{\pi i}+1=0$ without having to look at the formula editor or use the
kohlhase-omdoc_F00353	251	■ 0.370	$\{ \}^{\mathrm{TM}}$	TM	“%pi” for MAXIMA, or “Pi” for YACAS, MAPLE™ and MATHEMATICA®.
kohlhase-omdoc_F00354	251	■ 0.370	$\{ \}^{\mathrm{(R)}}$	(R)	“%pi” for MAXIMA, or “Pi” for YACAS, MAPLE™ and MATHEMATICA®.
kohlhase-omdoc_F00355	254	■ 1.000	0	O	servers (see [KF01] for details). In particular, an object O residing on an
kohlhase-omdoc_F00356	254	■ 1.000	$0^{\prime}$	O'	MBASE server S can refer to (or depend on) an object O' residing on a
kohlhase-omdoc_F00357	254	■ 1.000	$S^{\prime}$	S'	server S' ; a query to O that needs information about O' will be delegated to
kohlhase-omdoc_F00358	255	■ 1.000	$\binom{n}{k}, \{ \}_n C^k, C_{\{k\}}^n$	$\binom{n}{k}, {}_n C^k, C_k^n$	$\binom{n}{k}$, ${}_n C^k$, C_k^n , and C_n^k all mean the same thing: $\frac{n!}{k!(n-k)!}$. In a formula
kohlhase-omdoc_F00359	255	■ 1.000	$C_{\{n\}}^k$	C_n^k	$\binom{n}{k}$, ${}_n C^k$, C_k^n , and C_n^k all mean the same thing: $\frac{n!}{k!(n-k)!}$. In a formula
kohlhase-omdoc_F00360	255	■ 1.000	$\frac{n!}{k!(n-k)!}$	$\frac{n!}{k!(n-k)!}$	$\binom{n}{k}$, ${}_n C^k$, C_k^n , and C_n^k all mean the same thing: $\frac{n!}{k!(n-k)!}$. In a formula
kohlhase-omdoc_F00361	255	■ 0.999	$\int f(x) \, dx$	$\int f(x)dx$	<i>jects</i> , e.g. an integral expression of the form $\int f(x)dx$ can mean a Rie-
kohlhase-omdoc_F00362	255	■ 1.000	$\int f(y) \, dy$	$\int f(y)dy$	stance $\int f(x)dx$ means the same as $\int f(y)dy$ (modulo α -equivalence), so
kohlhase-omdoc_F00363	255	■ 1.000	$s(t)$	$s(t)$	cal problem such as finding the power of a given signal $s(t)$. Of course our
kohlhase-omdoc_F00364	255	■ 1.000	$\int_?^? s^2(t) \, dt$	$\int_?^? s^2(t)dt$	MATHWEBSEARCH to search for something that looks like $\int_?^? s^2(t)dt$ (for
kohlhase-omdoc_F00365	255	■ 1.000	$\frac{1}{T} \int_0^T \int_{-\infty}^{\infty} s^2(t) \, dt = \sum_{k=-\infty}^{\infty} c_k ^2$	$\frac{1}{T} \int_0^T s^2(t)dt = \sum_{k=-\infty}^{\infty} c_k ^2$	$\frac{1}{T} \int_0^T s^2(t)dt = \sum_{k=-\infty}^{\infty} c_k ^2$ where c_k are the Fourier coefficients of the
kohlhase-omdoc_F00366	255	■ 1.000	$c_{\{k\}}$	c_k	$\frac{1}{T} \int_0^T s^2(t)dt = \sum_{k=-\infty}^{\infty} c_k ^2$ where c_k are the Fourier coefficients of the
kohlhase-omdoc_F00367	257	■ 1.000	A=B	$A=B$	While of searching expressions of the form $A=B$, we might like to find
kohlhase-omdoc_F00368	257	■ 0.737	B=A	$B=A$	occurrences of $B=A$ as well. At this point the <code>mq:anyorder</code> attribute comes
kohlhase-omdoc_F00369	258	■ 0.994	$\frac{1}{T} \int_0^T \int_{-\infty}^{\infty} s^2(x) \, dx = \sum_{k=-\infty}^{\infty} c_k ^2$	$\frac{1}{T} \int_0^T s^2(x)dx = \sum_{k=-\infty}^{\infty} c_k ^2$	the variant $\frac{1}{T} \int_0^T s^2(x)dx = \sum_{k=-\infty}^{\infty} c_k ^2$.
kohlhase-omdoc_F00370	259	■ 0.319	$\left[\{ \}^+{}_{03}, \mathrm{KBLL} \{ \}^+{}_{04}, \mathrm{KBKB}^+{}_{04} \right]$	$^+{}_{03}, \mathrm{KBLL}^+{}_{04}, \mathrm{KBKB}^+{}_{04}$	(see [KBHL ⁺ 03, KBLL ⁺ 04, KBKB ⁺ 04]).
kohlhase-omdoc_F00371	261	■ 1.000	T	T	entities, such that (for a theorem T a proof P , and sections A, B):
kohlhase-omdoc_F00372	261	■ 1.000	A, B	A, B	entities, such that (for a theorem T a proof P , and sections A, B):
kohlhase-omdoc_F00373	261	■ 0.533	$P \ \& \ A$	$P\&A$	B contains P & A contains T & P proves $T \Rightarrow B$ reliesOn A .
kohlhase-omdoc_F00374	261	■ 0.533	$T \ \& \ P$	$T\&P$	B contains P & A contains T & P proves $T \Rightarrow B$ reliesOn A .
kohlhase-omdoc_F00375	261	■ 0.533	$T \Rightarrow B$	$T \Rightarrow B$	B contains P & A contains T & P proves $T \Rightarrow B$ reliesOn A .
kohlhase-omdoc_F00376	264	■ 0.997	$\mathbb{F}$	$\mathbb{F}$	ematical objects ‘live’. For instance, when a chapter begins with “Let $\mathbb{F}$ be
kohlhase-omdoc_F00377	264	■ 1.000	$\frac{3}{4} / \frac{2}{3} / \frac{3}{5}$	$\frac{3/4+2/3}{5}$	fraction display. In $\frac{3/4+2/3}{5}$ we have used them both.

Identifier	Page	Conf.	LaTeX source	Rendered	Image				
kohlhase-omdoc_F00378	264	0.998	<code>3,4 \in \mathbb{Z}</code>	$3,4 \in \mathbb{Z}$	$3,4 \in \mathbb{Z}$. The corresponding OPENMATH expression would be the equivalent				
kohlhase-omdoc_F00379	264	1.000	<code>3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}</code>	$3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$	of $3 \in \mathbb{Z} \wedge 4 \in \mathbb{Z}$, but our OPENMATH statement reads that the sequence				
kohlhase-omdoc_F00380	264	0.998	<code>\mathbb{Z}</code>	$\mathbb{Z}$	$3,4$ belongs to $\mathbb{Z}$. So here, the semantics of the element-of symbol has been				
kohlhase-omdoc_F00381	268	0.608	<code>\left[\mathrm{MGH}^{+}{}_{05}\right]</code>	$\left[\mathrm{MGH}^{+}{}_{05}\right]$	and tools [MGH ⁺ 05].				
kohlhase-omdoc_F00382	269	0.995	<code>\left[\mathrm{KAB}^{+}{}_{04}\right]</code>	$\left[\mathrm{KAB}^{+}{}_{04}\right]$	ment (PISA) [KAB ⁺ 04] and National Council of Teachers of Mathematics				
kohlhase-omdoc_F00383	273	0.996	<code>10^{\prime}000</code>	$10'000$	To date, more than 10'000 <i>items</i> of OMDoc content has been written				
kohlhase-omdoc_F00384	273	1.000	<code>1, \ldots, k, \ldots, n</code>	$1,\ldots,k,\ldots,n$	age of the ellipsis in such a formula as $1,\ldots,k,\ldots,n$ is a simple example of				
kohlhase-omdoc_F00385	275	0.998	<code>\left[\mathrm{SBB}^{+}{}_{06}\right]</code>	$\left[\mathrm{SBB}^{+}{}_{06}\right]$	learning [SBB ⁺ 06].				
kohlhase-omdoc_F00386	279	1.000	<code>{ }^{\sim}</code>	$\sim$	<table><tr><td>Project Home</td><td>http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html</td></tr><tr><td>Authors</td><td>Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis School of Computer Science and Information Technology, University of Nottingham</td></tr></table>	Project Home	http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html	Authors	Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis School of Computer Science and Information Technology, University of Nottingham
Project Home	http://www.cs.nott.ac.uk/~lad/research/challenges/challenge_manager.html								
Authors	Thomas D. Attfield, Monica C. Duarte, Lin Li, Ho-Ying Mak, Adam M. Neal, Lewis M. Toft, Zixuan Wang, Louise A. Dennis School of Computer Science and Information Technology, University of Nottingham								
kohlhase-omdoc_F00387	282	1.000	<code>{ }^{1} { }^{1}</code>	11	<table><tr><td>Project Home</td><td>www.dfki.de/~inka/maya.html</td></tr><tr><td>Authors</td><td>Serge Autexier¹, Dieter Hutter¹, Till Mossakowski², Axel Schairer¹ ¹ DKFI GmbH, Stuhlsatzenhausweg 3, D 66123 Saarbrücken ² Computer Science, University of Bremen, Germany</td></tr></table>	Project Home	www.dfki.de/~inka/maya.html	Authors	Serge Autexier ¹ , Dieter Hutter ¹ , Till Mossakowski ² , Axel Schairer ¹ ¹ DKFI GmbH, Stuhlsatzenhausweg 3, D 66123 Saarbrücken ² Computer Science, University of Bremen, Germany
Project Home	www.dfki.de/~inka/maya.html								
Authors	Serge Autexier ¹ , Dieter Hutter ¹ , Till Mossakowski ² , Axel Schairer ¹ ¹ DKFI GmbH, Stuhlsatzenhausweg 3, D 66123 Saarbrücken ² Computer Science, University of Bremen, Germany								
kohlhase-omdoc_F00388	282	0.757	<code>{ }^{+}{}_{00}</code>	$^{+}{}_{00}$	SL [AHL ⁺ 00]. All these specification languages provide constructs similar				
kohlhase-omdoc_F00389	286	0.830	<code>\left[\mathrm{SHB}^{+}{}_{99}\right]</code>	$\left[\mathrm{SHB}^{+}{}_{99}\right]$	of the GUI, shared with the Ω MEGA-system [SHB ⁺ 99], are written				
kohlhase-omdoc_F00390	295	0.465	<code>\mathrm{St}_{\mathrm{E}} \mathrm{X}</code>	$\mathrm{St}_{\mathrm{E}}\mathrm{X}$	We present $\mathfrak{S}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ (Semantic $\mathrm{T}_{\mathrm{E}}\mathrm{X}$) a collection of macro packages for				
kohlhase-omdoc_F00391	295	0.962	<code>\mathrm{ST}_{\mathrm{E}}^{\mathrm{X}} \mathrm{X}</code>	$\mathrm{ST}_{\mathrm{E}}^{\mathrm{X}}\mathrm{X}$	documents can be authored and maintained in $\mathfrak{S}\mathrm{T}_{\mathrm{E}}\mathrm{X}$ using a simple text ed-				
kohlhase-omdoc_F00392	295	0.580	<code>\mathrm{T}_{\mathrm{E}} \mathrm{X} / \mathrm{IAT}_{\mathrm{E}} \mathrm{X}</code>	$\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{IAT}_{\mathrm{E}}\mathrm{X}$	– The $\mathrm{T}_{\mathrm{E}}\mathrm{X} / \mathrm{IAT}_{\mathrm{E}}\mathrm{X}$ syntax is much more compact than OMDoc, and if				
kohlhase-omdoc_F00393	297	0.713	<code>\binom{n}{k}, { }_{n} C^{k}, \mathrm{C}_{k}^n</code>	$\binom{n}{k}, {}_nC^k, \mathcal{C}_k^n$	tations” ¹⁹ for binomial coefficients: $\binom{n}{k}$, ${}_nC^k$, $\mathcal{C}_k^n$, and $\mathcal{C}_n^k$ all mean the same				
kohlhase-omdoc_F00394	297	0.713	<code>\mathrm{C}_{n}^k</code>	$\mathcal{C}_n^k$	tations” ¹⁹ for binomial coefficients: $\binom{n}{k}$, ${}_nC^k$, $\mathcal{C}_k^n$, and $\mathcal{C}_n^k$ all mean the same				
kohlhase-omdoc_F00395	297	1.000	<code>\sum_{k=1}^{\infty} x^k</code>	$\sum_{k=1}^{\infty} x^k$	for the mathematical expression $\sum_{k=1}^{\infty} x^k$. In the first form, we specify that we				
kohlhase-omdoc_F00396	297	0.943	<code>\infty</code>	∞	(i) the bound variable k (ii) the number 1 (iii) ∞ (iv) $\backslash \mathrm{Cexp}\{\mathrm{x}\}k$ (i.e. x to				
kohlhase-omdoc_F00397	297	0.870	<code>k=1</code>	$k = 1$	capital sigma character (Σ) whose subscript is the equation $k = 1$ and whose				
kohlhase-omdoc_F00398	298	0.996	<code>\sum_{i=1}^n x^i</code>	$\sum_{i=1}^n x^i$	an expression like $\sum_{i=1}^n x^i$ as $\backslash \mathrm{Sumfromto}\{i\}1n\{2i-1\}$. In this example we				

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																								
kohlhase-omdoc_F00399	298	1.000	\arbitraryn	(not rendered)	What is the sum of the first $\$ \arbitraryn$ odd numbers, i.e.$																																																								
kohlhase-omdoc_F00400	298	1.000	\Sumfromto{i}1\arbitraryn{2i-1}?	(not rendered)	$\$ \Sumfromto{i}1 \arbitraryn {2i-1} ?$$																																																								
kohlhase-omdoc_F00401	298	0.998	\sum_{i=1}^n 2 i - 1	$\sum_{i=1}^n 2i - 1$	<u>What is the sum of the first n odd numbers, i.e. $\sum_{i=1}^n 2i - 1$?</u>																																																								
kohlhase-omdoc_F00402	298	0.660	\mathrm{S}_{\mathrm{E}}\mathrm{X}	S _E X	The g _E X macro packages have been validated together with a case																																																								
kohlhase-omdoc_F00403	303	0.999	a x^{2}+b x+c	$ax^2 + bx + c$	mial expression $ax^2 + bx + c$ whose internal representation in MATHEMATICA [®]																																																								
kohlhase-omdoc_F00404	306	0.994	\Omega_{\text {MEGA }}	Ω _{MEGA}	<table><tr><td>PVS</td><td>Ω_{MEGA}</td><td>PVS</td><td>Ω_{MEGA}</td></tr><tr><td>set</td><td></td><td>subset?</td><td>subset</td></tr><tr><td>member</td><td>in</td><td></td><td>subset2</td></tr><tr><td>empty?</td><td>empty</td><td>strict_subset?</td><td>proper-subset</td></tr><tr><td>emptyset</td><td>emptyset</td><td></td><td>superset</td></tr><tr><td>nonempty?</td><td>not-empty</td><td>union</td><td>union</td></tr><tr><td>full?</td><td></td><td></td><td>union2</td></tr><tr><td>fullset</td><td></td><td></td><td>union-over-collection</td></tr><tr><td>singleton?</td><td>singleton</td><td>intersection</td><td>intersection</td></tr><tr><td>singleton</td><td></td><td></td><td>intersection-over-coll.</td></tr><tr><td>complement</td><td>set-complement</td><td>disjoint?</td><td>misses</td></tr><tr><td>difference</td><td>setminus</td><td>meets</td><td></td></tr><tr><td>symmetric.difference</td><td></td><td>add</td><td>add-one</td></tr><tr><td></td><td>exclunion</td><td>remove</td><td></td></tr></table>	PVS	Ω _{MEGA}	PVS	Ω _{MEGA}	set		subset?	subset	member	in		subset2	empty?	empty	strict_subset?	proper-subset	emptyset	emptyset		superset	nonempty?	not-empty	union	union	full?			union2	fullset			union-over-collection	singleton?	singleton	intersection	intersection	singleton			intersection-over-coll.	complement	set-complement	disjoint?	misses	difference	setminus	meets		symmetric.difference		add	add-one		exclunion	remove	
PVS	Ω _{MEGA}	PVS	Ω _{MEGA}																																																										
set		subset?	subset																																																										
member	in		subset2																																																										
empty?	empty	strict_subset?	proper-subset																																																										
emptyset	emptyset		superset																																																										
nonempty?	not-empty	union	union																																																										
full?			union2																																																										
fullset			union-over-collection																																																										
singleton?	singleton	intersection	intersection																																																										
singleton			intersection-over-coll.																																																										
complement	set-complement	disjoint?	misses																																																										
difference	setminus	meets																																																											
symmetric.difference		add	add-one																																																										
	exclunion	remove																																																											
kohlhase-omdoc_F00405	306	1.000	\rho_{*}	ρ _*	inclusions ρ _* that interpret the system concepts in terms of mbase:sets.																																																								
kohlhase-omdoc_F00406	306	1.000	\rho_{*}^{-1}	ρ _* ^{−1}	the integration theory, using the partial inverse ρ _* ^{−1} of ρ _* . For an integration																																																								
kohlhase-omdoc_F00407	308	1.000	\mathcal{S}=(\mathcal{L},\mathcal{C})	$\mathcal{S} = (\mathcal{L}, \mathcal{C})$	Definition 26.18.1. A <i>logical system</i> $\mathcal{S} = (\mathcal{L}, \mathcal{C})$ consists of a language $\mathcal{L}$																																																								
kohlhase-omdoc_F00408	308	1.000	\mathcal{L}	$\mathcal{L}$	Definition 26.18.1. A <i>logical system</i> $\mathcal{S} = (\mathcal{L}, \mathcal{C})$ consists of a language $\mathcal{L}$																																																								
kohlhase-omdoc_F00409	308	0.999	\mathcal{H}	$\mathcal{H}$	rules). A calculus gives us a notion of a $\mathcal{C}$ -derivation of A from $\mathcal{H}$, which																																																								
kohlhase-omdoc_F00410	309	1.000	\mathcal{D}: \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}	$\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$	we will denote by $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$. Let $\mathcal{S}$ and $\mathcal{S}'$ be logical systems, then a <i>logic</i>																																																								
kohlhase-omdoc_F00411	309	1.000	\mathcal{S}^{\prime}	$\mathcal{S}'$	we will denote by $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$. Let $\mathcal{S}$ and $\mathcal{S}'$ be logical systems, then a <i>logic</i>																																																								
kohlhase-omdoc_F00412	309	1.000	\mathcal{F}: \mathcal{S} \rightarrow \mathcal{S}^{\prime}	$\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}'$	<i>morphism</i> $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}'$ consists of a <i>language morphism</i> $\mathcal{F}^{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{L}'$ and																																																								
kohlhase-omdoc_F00413	309	1.000	\mathcal{F}^{\mathcal{L}}: \mathcal{L} \rightarrow \mathcal{L}^{\prime}	$\mathcal{F}^{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{L}'$	<i>morphism</i> $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}'$ consists of a <i>language morphism</i> $\mathcal{F}^{\mathcal{L}} : \mathcal{L} \rightarrow \mathcal{L}'$ and																																																								
kohlhase-omdoc_F00414	309	1.000	\mathcal{F}^{\mathcal{D}}	$\mathcal{F}^{\mathcal{D}}$	a <i>calculus morphism</i> $\mathcal{F}^{\mathcal{D}}$ from $\mathcal{C}$ -derivations to $\mathcal{C}'$ -derivations, such that																																																								
kohlhase-omdoc_F00415	309	1.000	\mathcal{C}^{\prime}	$\mathcal{C}'$	a <i>calculus morphism</i> $\mathcal{F}^{\mathcal{D}}$ from $\mathcal{C}$ -derivations to $\mathcal{C}'$ -derivations, such that																																																								
kohlhase-omdoc_F00416	309	0.999	\mathcal{F}^{\mathcal{D}}(\mathcal{D}): \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}'} \mathcal{F}^{\mathcal{L}}(\mathbf{A})	$\mathcal{F}^{\mathcal{D}}(\mathcal{D}) : \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}'} \mathcal{F}^{\mathcal{L}}(\mathbf{A})$	for any $\mathcal{C}$ -derivation $\mathcal{D} : \mathcal{H} \vdash_{\mathcal{C}} \mathbf{A}$ we have $\mathcal{F}^{\mathcal{D}}(\mathcal{D}) : \mathcal{F}^{\mathcal{L}}(\mathcal{H}) \vdash_{\mathcal{C}'} \mathcal{F}^{\mathcal{L}}(\mathbf{A})$.																																																								
kohlhase-omdoc_F00417	309	1.000	\mathcal{R}	$\mathcal{R}$	instance the morphism $\mathcal{R}$ from sorted first-order logic (<i>SFOL</i>) to unsorted																																																								
kohlhase-omdoc_F00418	309	1.000	(\mathbb{S} F O L)	(<i>SFOL</i>)	instance the morphism $\mathcal{R}$ from sorted first-order logic (<i>SFOL</i>) to unsorted																																																								
kohlhase-omdoc_F00419	309	1.000	(F O L)	(<i>FOL</i>)	first-order logic (<i>FOL</i>). For every sorted constant $\mathcal{R}$ introduces an axiom																																																								
kohlhase-omdoc_F00420	309	1.000	\mathcal{R}([+: \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}]) = \forall X, Y . \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X + Y)	$\mathcal{R}([+ : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}]) = \forall X, Y. \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X + Y)$	e.g. $\mathcal{R}([+ : \mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}]) = \forall X, Y. \mathbb{N}(X) \wedge \mathbb{N}(Y) \Rightarrow \mathbb{N}(X + Y)$. On formu-																																																								

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_ F00421	309	■ 0.999	$\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A}) = \forall X.\mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$ $\right)=\forall\text{forall } X . \mathbb{B}(X) \rightarrow$ $\mathcal{R}(\mathbf{A})$	$\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A}) = \forall X.\mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$	predicates, e.g. $\mathcal{R}(\forall X_{\mathbb{B}}.\mathbf{A}) = \forall X.\mathbb{B}(X) \Rightarrow \mathcal{R}(\mathbf{A})$. Finally, for proofs we have
kohlhase-omdoc_ F00422	309	■ 1.000	$\mathbb{A}, \mathbb{B}, \dots$	$\mathbb{A}, \mathbb{B}, \dots$	the correspondence given in Figure 26.26, where $\mathbb{A}, \mathbb{B}, \dots$ are sort symbols.
kohlhase-omdoc_ F00423	309	■ 0.989	$\mathbb{S} \text{ F O L}$	$\mathcal{S}FOL$	simply be encoded as a theory inclusion from $\mathcal{S}FOL$ to FOL mapping $\mathcal{S}FOL$
kohlhase-omdoc_ F00424	309	■ 0.989	F O L	FOL	simply be encoded as a theory inclusion from $\mathcal{S}FOL$ to FOL mapping $\mathcal{S}FOL$
kohlhase-omdoc_ F00425	310	■ 0.752	$\left[\mathrm{SBB}^{+02}\right]$	$\left[\mathrm{SBB}^{+02}\right]$	ΩMEGA [SBB ⁺ 02, SBA05]. Generally, we aim at an approach that is not de-
kohlhase-omdoc_ F00426	310	■ 0.975	$\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}[\mathrm{dH}01]$	$\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}[\mathrm{dH}01]$	$\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}[\mathrm{dH}01]$ is a scientific WYSIWYG text editor that provides
kohlhase-omdoc_ F00427	312	■ 1.000	$T+S$	$T+S$	we define a semantic document format $T+S$ as a document format still
kohlhase-omdoc_ F00428	312	■ 1.000	$S(P)$	$S(P)$	defining a fixed language S , we define a language $S(P)$ parametrized over a
kohlhase-omdoc_ F00429	312	■ 1.000	$S(C)$	$S(C)$	language P for proofs and define the document format based on $S(C)$, where
kohlhase-omdoc_ F00430	312	■ 1.000	C	C	C denotes the proof language of CORE. This format supports the <i>static repre-</i>
kohlhase-omdoc_ F00431	312	■ 0.993	$T+S(C)$	$T+S(C)$	The $\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}$ document $T+S(C)$ and the pure semantic representation
kohlhase-omdoc_ F00432	312	■ 0.993	t_{s_i}	ts_i	$\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}$ documents. The differences between two versions ts_i and ts_{i+1} of
kohlhase-omdoc_ F00433	312	■ 0.993	$t_{s_{i+1}}$	ts_{i+1}	$\mathrm{T}_{\mathrm{E}}\mathrm{X}_{\mathrm{MACS}}$ documents. The differences between two versions ts_i and ts_{i+1} of
kohlhase-omdoc_ F00434	312	■ 1.000	s_{i_i}	s_i	corresponding document s_i in $S(C)$ for ts_i , such that the application of p
kohlhase-omdoc_ F00435	312	■ 1.000	s_{i+1}	s_{i+1}	to s_i results in s_{i+1} which corresponds to ts_{i+1} . An analogous diff/patch
kohlhase-omdoc_ F00436	313	■ 1.000	$T+S(C)+M$	$T+S(C)+M$	extended to $T+S(C)+M$, where the menus can be attached to arbitrary
kohlhase-omdoc_ F00437	313	■ 1.000	$S(C)+M$	$S(C)+M$	and $S(C)+M$ via the diff/patch mechanism. Note that this includes also
kohlhase-omdoc_ F00438	314	■ 1.000	$\mathcal{F}\mathcal{P}$	$\mathcal{F}\mathcal{P}$	gramming language $\mathcal{F}\mathcal{P}$. The system has been developed since 1998 at
kohlhase-omdoc_ F00439	316	■ 0.537	$h_1, \dots h_n, \forall \dots i h_1, \dots, \forall \dots i h_l \vdash$ $\forall \dots i h_{\{l\}} \vdash$	$h_1, \dots h_n, \forall \dots i h_1, \dots, \forall \dots i h_l \vdash$	with a sequence of the form $h_1, \dots h_n, \forall \dots i h_1, \dots, \forall \dots i h_l \vdash goal$ whereas
kohlhase-omdoc_ F00440	316	■ 1.000	h_{i_i}	h_i	the expressions h_i are the hypotheses, the expressions $\forall \dots i h_k$ are the in-
kohlhase-omdoc_ F00441	316	■ 1.000	$\forall \dots i h_{\{k\}}$	$\forall \dots i h_k$	the expressions h_i are the hypotheses, the expressions $\forall \dots i h_k$ are the in-
kohlhase-omdoc_ F00442	332	—	$\begin{aligned} & \& ([0-9] , \cdot) \& (* \mid \\ & \mathrm{e}([0-9] , \cdot) *) ? \end{aligned}$	$([0-9] , \cdot)$ $(* \mid \mathrm{e}([0-9] , \cdot)*)?$	—

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kohlhase-omdoc_ F00443	338	0.805	f(a, b)	$f(a, b)$	<table><tr><th>Attribute</th><th>element</th><th>Values</th></tr><tr><td>action</td><td>dc:date</td><td>unspecified</td></tr><tr><td></td><td colspan="2">specifies the action taken on the document on this date.</td></tr><tr><td>action</td><td>omlet</td><td>execute, display, other</td></tr><tr><td></td><td colspan="2">specifies the action to be taken when executing the omlet, the value is application-defined.</td></tr><tr><td>actuate</td><td>omlet</td><td>onPresent, onLoad, onRequest, other</td></tr><tr><td></td><td colspan="2">specifies the timing of the action specified in the action attribute</td></tr><tr><td>assertion</td><td>example</td><td></td></tr><tr><td></td><td colspan="2">specifies the assertion that states that the objects given in the example really have the expected properties.</td></tr><tr><td>assertion</td><td>obligation</td><td></td></tr><tr><td></td><td colspan="2">specifies the assertion that states that the translation of the statement in the source theory specified by the induced-by attribute is valid in the target theory.</td></tr><tr><td>attributes</td><td>use</td><td></td></tr><tr><td></td><td colspan="2">the attribute string for the start tag of the XML element substituted for the brackets (this is specified in the element attribute).</td></tr><tr><td>attribution</td><td>cc:requirements</td><td>required, not_required</td></tr><tr><td></td><td colspan="2">Specifies whether the copyright holder/author must be given credit in derivative works</td></tr><tr><td>base</td><td>morphism</td><td></td></tr><tr><td></td><td colspan="2">specifies another morphism that should be used as a base for expansion in the definition of this morphism</td></tr><tr><td>bracket-style</td><td>presentation, use</td><td>lisp, math</td></tr><tr><td></td><td colspan="2">specifies whether a function application is of the form $f(a, b)$ or (fab)</td></tr><tr><td>cd</td><td>om:OMS</td><td></td></tr><tr><td></td><td colspan="2">specifies the content dictionary of an OPENMATH symbol</td></tr><tr><td>cd</td><td>term</td><td></td></tr><tr><td></td><td colspan="2">specifies the content dictionary of a technical term</td></tr><tr><td>cdbase</td><td>om:*</td><td></td></tr><tr><td></td><td colspan="2">specifies the base URI of the content dictionaries used in an OPENMATH object</td></tr><tr><td>cdreviewdate</td><td>theory</td><td></td></tr><tr><td></td><td colspan="2">specifies the date until which the content dictionary will remain unchanged</td></tr><tr><td>cdrevision</td><td>theory</td><td></td></tr><tr><td></td><td colspan="2">specifies the minor version number of the content dictionary</td></tr><tr><td>cdstatus</td><td>theory</td><td>official, experimental, private, obsolete</td></tr><tr><td></td><td colspan="2">specifies the content dictionary status</td></tr><tr><td>cdurl</td><td>theory</td><td></td></tr><tr><td></td><td colspan="2">the main URL, where the newest version of the content dictionary can be found</td></tr></table>	Attribute	element	Values	action	dc:date	unspecified		specifies the action taken on the document on this date.		action	omlet	execute, display, other		specifies the action to be taken when executing the omlet, the value is application-defined.		actuate	omlet	onPresent, onLoad, onRequest, other		specifies the timing of the action specified in the action attribute		assertion	example			specifies the assertion that states that the objects given in the example really have the expected properties.		assertion	obligation			specifies the assertion that states that the translation of the statement in the source theory specified by the induced-by attribute is valid in the target theory.		attributes	use			the attribute string for the start tag of the XML element substituted for the brackets (this is specified in the element attribute).		attribution	cc:requirements	required, not_required		Specifies whether the copyright holder/author must be given credit in derivative works		base	morphism			specifies another morphism that should be used as a base for expansion in the definition of this morphism		bracket-style	presentation, use	lisp, math		specifies whether a function application is of the form $f(a, b)$ or (fab)		cd	om:OMS			specifies the content dictionary of an OPENMATH symbol		cd	term			specifies the content dictionary of a technical term		cdbase	om:*			specifies the base URI of the content dictionaries used in an OPENMATH object		cdreviewdate	theory			specifies the date until which the content dictionary will remain unchanged		cdrevision	theory			specifies the minor version number of the content dictionary		cdstatus	theory	official, experimental, private, obsolete		specifies the content dictionary status		cdurl	theory			the main URL, where the newest version of the content dictionary can be found	
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kohlhase-omdoc_F00445	374	0.966	{ }^{+} 02	$^{+}02$	[BCD ⁺ 02] R. Bradford, R. M. Corless, J. H. Davenport, D. J. Jeffrey, and S. M.																																																																																																			
kohlhase-omdoc_F00446	375	1.000	\bar{\lambda} \mu \bar{\mu}	$\bar{\lambda}\mu\bar{\mu}$	[Coe05] Claudio Sacerdoti Coen. Explanation in natural language of $\bar{\lambda}\mu\bar{\mu}$ -terms.																																																																																																			
kohlhase-omdoc_F00447	377	0.891	H O L-A	$HOL - A$	[GM93] M. J. C. Gordon and T. F. Melham. <i>Introduction to HOL – A theorem</i>																																																																																																			
kohlhase-omdoc_F00448	379	1.000	T_{E} X b o o k	T_EXbook	[Knu84] Donald E. Knuth. <i>The T_EXbook</i> . Addison Wesley, 1984.																																																																																																			
kohlhase-omdoc_F00449	380	0.807	{ }^{A T} T_{E} X	$^{AT}T_EX$	[Lam94] Leslie Lamport. <i>ΛT_EX: A Document Preparation System, 2/e</i> . Addison																																																																																																			
kohlhase-omdoc_F00450	383	0.842	\lambda c l a m	$\lambda clam$	scription: Proof planning in higher-order logic with $\lambda clam$. In Claude Kirchner																																																																																																			
kohlhase-omdoc_F00451	383	0.990	{ }^{+} 06	$^{+}06$	[SBB ⁺ 06] Sebastian Schaffert, Diana Bischof, Tobias Bürger, Andreas Gruber,																																																																																																			
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kohlhase-omdoc_F00453	384	0.614	\mathcal{L} \mathcal{O} \mathcal{U} \mathcal{I}	$\mathcal{LOUI}$	Konrad, Andreas Meier, Erica Melis, Martin Pollet, and Volker Sorge. $\mathcal{LOUI}$:																																																																																																			
kohlhase-omdoc_F00454	384	0.952	\mathcal{U}	$\mathcal{U}$	Lovely Ω MEGA User Interface. <i>Formal Aspects of Computing</i> , 11:326–342, 1999.																																																																																																			
kohlhase-omdoc_F00455	384	0.952	\mathcal{I}	$\mathcal{I}$	Lovely Ω MEGA User Interface. <i>Formal Aspects of Computing</i> , 11:326–342, 1999.																																																																																																			
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